

Master Thesis

Self-Organization Patterns in LLM-based Multi-Agent Systems

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Abstract

A popular way to improve how language models reason is multi-agent debate, in which several agents discuss a problem over multiple rounds. As such systems move into real applications, it matters not only whether they reach good answers but also whether the way they reach them can be trusted. Recent studies raise doubts on both counts, showing that debate often does no better than a simple vote, as groups agree too early on what they already share and agents abandon correct answers to follow confident peers. These failures arise from how the discussion unfolds, yet debate is mostly judged by outcome measures such as final accuracy. Such measures cannot tell apart two debates that reach the same answer, one because the group corrected an early mistake and the other because it simply amplified the answer most of its agents started with. Studying the discussion itself calls for a theory of how interacting parts produce collective behaviour, which is exactly what complex systems theory offers. If debate behaves like other complex systems, its failures should leave recognisable patterns that explain why it goes wrong and could signal problems early. We connect this theory to multi-agent debate and introduce Self-Organization Motif Analysis (SOMA), a method that detects four such patterns in debates, from self-reinforcing positions to emerging leaders.

Applied to 30,000 debates, these patterns reveal a single mechanism, as debate mostly amplifies the answers agents hold at the start rather than producing new ones. When we change only how many agents start with the correct answer, for instance, the group ends correctly more often with each additional correct agent, while it rarely reaches a correct answer that none of its agents started with. Diverse starting answers, in turn, make it more likely that the group follows an agent who was right from the beginning. Since SOMA largely works with information that many debate systems already record, the path to an answer becomes something practitioners can measure rather than merely assume. This thesis thus connects complex systems theory with multi-agent debate and turns the resulting insights into concrete design guidance.

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1 Introduction

1.1 Motivation

Large language models (LLMs) are increasingly deployed not as isolated predictors but as interacting agents that reason, argue, and decide in groups. A prominent instance of this shift is *multi-agent debate* (MAD), in which several LLM instances independently propose answers, exchange their reasoning over multiple rounds, and revise their positions before the group commits to a final decision (Du et al., 2024). Across factual and mathematical reasoning tasks, this simple protocol has been reported to improve accuracy and reduce hallucination relative to a single model answering alone (Du et al., 2024, Liang et al., 2024, Chen, Saha and Bansal, 2024), and it now underpins a growing family of collaborative and evaluative systems (Chan et al., 2024). The appeal is intuitive: independent agents make partially independent errors, and deliberation offers a mechanism for the group to identify and correct them.

This optimistic reading, however, rests almost entirely on *outcome* measurements. The literature evaluates multi-agent systems predominantly by final task performance, treating the deliberation itself as an opaque process that either does or does not yield a better answer. Recent critical reassessments show that this outcome-centric view is fragile: debate frequently fails to outperform single-model baselines, collapses under controlled comparison with simpler alternatives (Zhang et al., 2025, Choi et al., 2025, Bertalaníč and Fortuna, 2026); multi-agent LLM systems more broadly exhibit failure rates between 41% and 86.7% across seven frameworks (Cemri et al., 2026). More troublingly, the failures that do occur are not random noise but exhibit structure. When two LLMs debate, both tend to grow *more* confident that they have won as the exchange proceeds, with confidence escalating away from a rational baseline even when no new supporting evidence is introduced (Prasad and Nguyen, 2025). Under distributed information, groups of LLM agents converge prematurely on the evidence they already share and fail to surface the private knowledge needed for the correct answer, so that collective accuracy collapses far below what a single agent achieves with complete information (?). Purely cooperative framings can further collapse the debate into a mutually agreeing echo chamber.

What these pathologies have in common is that they are *dynamical*. Rising confidence, premature lock-in, and conformity cascades are properties of how positions evolve and influence one another across rounds, not of any single agent’s competence at a single point in time. An outcome metric, by construction, cannot see them: two systems can reach the same final accuracy through entirely different trajectories, one through genuine error correction and the other through a confident slide into a shared mistake. Understanding *when* and *why* multi-agent deliberation succeeds therefore requires a vocabulary for the interaction dynamics themselves. Miehl et al. (2025) make precisely this argument at a conceptual level, contending that agentic AI must be studied through a systems-theoretic lens because emergent group-level behaviour cannot be predicted from individual agent properties alone. Their position, however, remains programmatic and offers no empirical instrument for detecting such behaviour in practice.

Complex-systems theory supplies exactly such an instrument. Systems of many interacting units that produce ordered collective behaviour without central control, the defining feature of self-organization (Ashby, 1947), are known to fall into a small number of recurring dynamical regimes. These regimes have precise names and precise signatures: *self-reinforcement*, the positive feedback by which a state amplifies its own persistence (Arthur, 1989); *limit cycles*, self-sustained periodic oscillations that never settle (Strogatz, 2015); *multistability*, the coexistence of multiple stable outcomes selected by initial conditions (Feudel, 2008); and *dominance*, the concentration of influence onto a single node (DeGroot, 1974, Jia et al., 2015). Such patterns are studied as *motifs*, recurring structural or dynamical building blocks whose presence in a system is both detectable and diagnostic of its behaviour (Milo et al., 2002, Alon, 2007, Driscoll et al., 2024). The pathologies documented in the MAD literature read naturally as instances of this vocabulary: escalating confidence is a self-reinforcement signature, premature consensus is a fixed-point attractor reached through basin selection, and herding to a confident leader is a dominance motif. The connection has begun to be made. Ashery et al. (2025) demonstrate the spontaneous emergence of universally adopted social conventions in decentralised LLM populations, show that strong collective biases arise even when individual agents exhibit none, and find that committed adversarial minorities can impose alternative conventions on the majority. Okawa (2026) derives a formal model predicting that once conformity pressure among agents exceeds a critical threshold, even small individual biases are amplified into a shared collective bias. What these convention-based treatments share is a substrate in which no option is better than any other, so they can show that a group locks in but not whether it locks in on the right answer; the epistemic case, where some claims are better supported than others, remains open.

This thesis addresses that gap. We construct a controlled cooperative multi-agent debate system in which homogeneous LLM agents deliberate over multiple rounds on tasks with verifiable ground truth, and we develop a suite of detectors, each grounded in the complex-systems and opinion-dynamics literature, that identify self-reinforcement, limit cycles, multistability, and dominance in the resulting interaction logs. We run this system across a controlled grid of communication topologies and memory configurations on two complementary benchmarks: GPQA-Diamond (Rein et al., 2024), a graduate-level science question-answering benchmark where agents typically start with reasonable priors, and HiddenBench (?), a hidden-profile benchmark constructed so that agents start systematically wrong and success requires genuine information pooling. The contrast between the two isolates whether observed dynamics depend on the presence of a reliable individual signal or persist even when the group must correct a shared bias. Throughout, we treat motif detection not as a competitor to accuracy but as an explanatory layer beneath it: a way to characterise the trajectory that produces an outcome, and thereby to understand which design choices shape the deliberative dynamics of LLM collectives.

This explanatory layer has a practical purpose. As multi-agent systems are deployed in consequential settings, an operator needs to know not only whether a configuration tends to produce correct answers but whether a given deliberation is proceeding soundly or pathologically, whether the group is genuinely correcting an error or confidently converging on a shared mistake, and whether it will terminate at all. Outcome metrics cannot answer

these questions. Instruments are beginning to appear from other directions: learned classifiers that predict from interpretable features whether a query warrants debate at all (Fan et al., 2026). What does not yet exist is an instrument grounded in the dynamical vocabulary itself, one that names which regime a deliberation is in rather than only flagging that something is wrong. A central thread of this thesis is therefore that each motif detector, being computed from ordinary interaction logs, doubles as a candidate *diagnostic* of deliberative health. We do not deliver a validated diagnostic tool; we deliver the empirical groundwork for one, a set of measurable signatures, characterised across a controlled grid, together with tiered, evidence-graded design principles that state honestly how far each is supported by our data.

In one sentence, the objective of this thesis is to understand *how* a cooperative LLM debate arrives at its answer, by measuring the self-organization patterns in its trajectory, and to turn that understanding into evidence-graded guidance for building such systems.

1.2 Research Questions

The overarching goal of this thesis is to establish whether the dynamical vocabulary of complex systems can be applied empirically to LLM-based multi-agent debate, and to characterise the conditions under which self-organization patterns emerge. We decompose this goal into three research questions, ordered by the class of evidence each requires.

- RQ1 Measurement.** Can self-organization motifs, specifically self-reinforcement, limit cycles, multistability, and dominance, be detected in cooperative LLM-based multi-agent debate using detectors grounded in the complex-systems and opinion-dynamics literature and each tested against a principled null model? And which of these motifs actually occur, at what prevalence, and with what structure?
- RQ2 Characterisation.** What governs the presence and strength of each motif, and what do they predict about system outcomes? We examine the two canonical structural axes, communication topology and memory window, together with the information regime of the task and the composition of the opening votes, and ask what each motif implies for accuracy, convergence speed, and worst-case runtime.
- RQ3 Causation.** Is the composition of the opening votes causally responsible for the outcome? Can the system generate a correct answer that no agent initially held, or does deliberation function primarily as an amplifier of the initial vote distribution?

One commitment underpins these questions. Every detector is tested against a null model that destroys the pattern we look for and leaves everything else intact. This matters because our state space is small: with four agents and three or four options, votes repeat and influence concentrates by chance alone. The null tells us what chance looks like, so a near-zero prevalence becomes a measurement rather than a failure. We report the motifs we do not find alongside those we do.

1.3 Contributions & Structure

This thesis makes three contributions, one for each research question.

- **An instrument for measuring debate dynamics.** We define a two-phase synchronous debate protocol that separates public messages from private beliefs and confidences, so that the interaction log exposes exactly the quantities motif detection requires (Section 4). On top of it we specify SOMA (Self-Organization Motif Analysis): detectors for self-reinforcement, multistability, dominance, and limit cycles, each grounded in the complex-systems or opinion-dynamics literature and each tested against its own null model (Section 5). Every detector states what it can and cannot claim, distinguishing a measured signature from a certified attractor. Running $R = 50$ repetitions of every task-config cell is what makes across-repetition attractor statistics measurable at all.
- **An empirical characterisation at scale.** We apply SOMA to 30,000 debates across twelve configurations and two deliberately contrasting benchmarks: a refinement regime in which agents mostly start correct, and a correction regime in which they start systematically wrong (Section 6). Self-reinforcement is universal and drives convergence, multistability is common and is a stable property of the task rather than of the configuration, dominance emerges behaviourally even under a symmetric topology, and genuine limit cycles are essentially absent outside the star, where they arise through a hub-driven mechanism. Reading the emergent influence graph within fully-connected runs reveals a lever the structural topology parameter cannot reach: a rare effective star, formed preferentially from diverse opening positions, is markedly more accurate than the fully-connected mean. Throughout, the structural axes we varied move the dynamics but not the accuracy.
- **A causal test of the initial-condition effect.** A seeded-initialisation experiment manipulates only the composition of the opening votes, leaving the debate itself unchanged. Inserting a single correct agent into an otherwise-incorrect opening raises accuracy from 1.8 % to 26.2 %, and does so in all 50 tasks tested. In the unseeded arm, where no agent starts correct, the system arrives at the correct answer in under 2 % of repetitions, and its ability to do so declines as tasks get harder. A devil’s advocate probe confirms the complementary vulnerability under a controlled composition: a single agent that never updates its position outweighs two agents that start correct, collapsing accuracy from 63.7 % to 10.0 %. Together these upgrade the central claim from association to intervention: within the setting we tested, deliberation functions predominantly as a selective amplifier of its opening vote distribution (Section 6).

From these results we derive design guidance graded by how strongly our data support each claim, and show that every SOMA detector, computed from ordinary interaction logs, is a candidate observable for monitoring deliberative health (Section 7). We deliver the measurements and the mapping from principle to observable, not a packaged tool.

Structure of this thesis. The thesis follows the arc of the three research questions. *Background* (Section 2) introduces the vocabulary of self-organization and dynamical motifs and fixes what we mean by an LLM agent and a debate; *Related Work* (Section 3) surveys how multi-agent systems are currently analysed and locates the gap this thesis addresses. The instrument is then built in two steps: *System and Experimental Setup* (Section 4) specifies the debate protocol, the two contrasting benchmarks, and the experimental grid, and *Detection Methodology* (Section 5) defines the four SOMA detectors together with the null model each is tested against. *Results* (Section 6) applies the instrument, reporting accuracy and efficiency, the four motifs in turn, the emergent influence topology, and the seeded-initialisation experiment. *Discussion* (Section 7) reads the motifs together as one amplifier mechanism and distils the findings into evidence-graded design principles. *Limitations and Future Work* (Section 7.4) states the scope of the claims and the agenda they imply, and *Conclusion* (Section 8) returns to the three research questions.

2 Background

This section provides the conceptual vocabulary needed to follow the rest of the thesis. Section 2.1 introduces complex systems, self-organization, and dynamical systems as the formal language for describing collective behaviour, including the long-run regimes a system can settle into. Section 2.2 introduces motifs and the opinion-dynamics models of belief updating under repeated interaction. Section 2.3 fixes what is meant by an LLM agent and by multi-agent debate. The exact motifs this thesis detects, and the methodology used to detect them, are defined in Section 5.1; this section only fixes the vocabulary they build on.

2.1 Complex Systems and Self-Organization

A *complex system* is a collection of many interacting components whose collective behaviour cannot be inferred from the properties of the individual components alone (Anderson, 1972, Ladyman et al., 2013). The behaviour is called *emergent*: a stock-market bubble, for instance, is produced by millions of traders acting on local information with no coordinating authority, yet the resulting price dynamic is not present in any individual trade (Mitchell, 2009). When such emergent order arises without any central controller, the system is called *self-organizing* (Ashby, 1947, Camazine et al., 2001). There are four ingredients that recur across self-organizing systems (Camazine et al., 2001): (i) *positive feedback*, which amplifies small deviations and drives the system away from uniformity; (ii) *negative feedback*, which prevents runaway growth and bounds the system within reachable states; (iii) *fluctuations*, the stochastic perturbations that allow the system to explore the state space; and (iv) *repeated local interactions*, through which global structure accumulates over many iterations from purely local rules. The organising principle is encoded entirely in these interactions, not in any single component and not in any external authority. Because the same local rule can lead to different long-run outcomes depending on which fluctuations are realised and where the system starts, the interaction rule fixes the *repertoire* of possible attractors while the initial configuration and the realised noise jointly select which one is reached.

To study self-organizing systems rigorously, one needs a language for how states evolve over time. That language is *dynamical systems theory* (Strogatz, 2015). The most general deterministic form is the discrete-time map

$$\mathbf{x}(t+1) = F(\mathbf{x}(t)), \quad (1)$$

where $\mathbf{x}(t)$ is the full state at step t and F encodes the interaction law. The space of all possible states is the *phase space*; as time advances, the system traces a *trajectory* through it. Three extensions cover practically important cases. In *stochastic dynamics* the update is a transition kernel, $\mathbf{x}(t+1) \sim P(\cdot | \mathbf{x}(t))$, and the trajectory is a Markov chain; absorbing states and recurrent classes then replace fixed points and cycles as the fundamental long-run objects. In *memory systems* the rule looks back k steps, $\mathbf{x}(t+1) =$

$F(\mathbf{x}(t), \dots, \mathbf{x}(t - k + 1))$, so the effective state is the window of the last k observations. In *discrete-state systems* $\mathbf{x}(t)$ takes values in a finite or countable set, and update rules describe transition tables rather than differential equations. The debate systems studied in this thesis instantiate all three at once: the state is discrete (each agent casts one of finitely many votes), the update is stochastic (sampling temperature), and it carries bounded memory (each agent conditions on a finite window of recent rounds).

The central question in dynamical systems theory is where trajectories end up. Most trajectories, after an initial transient, settle toward specific regions called *attractors*: invariant sets to which nearby trajectories converge. The classical image is a ball rolling under gravity on a hilly landscape: wherever the ball starts, friction brings it to rest in one of the valleys. However, not every fixed set is an attractor; for example, a ball balanced on a hilltop is also an equilibrium but it is unstable, because any small perturbation moves it away rather than back. An attractor is a stable fixed set on which small deviations decay. The *basin of attraction* is the set of initial conditions whose trajectories converge to that attractor; its *depth* is the barrier height that must be overcome to escape, while its *width* is the volume of initial conditions it captures. Depth and width are independent: a deep but narrow well is easy to stay in once entered but hard to reach from most starting points. The simplest and most common dynamical regime is *fixed-point convergence*: the system settles to a state $\mathbf{x}^*$ satisfying $F(\mathbf{x}^*) = \mathbf{x}^*$ and stays there. A structurally different regime is the *limit cycle*: a closed periodic orbit to which nearby trajectories spiral, so the system revisits the same sequence of states indefinitely without ever settling. Unlike the neutrally stable oscillation of a frictionless pendulum, a limit cycle is attracting (asymptotically stable): small perturbations push the trajectory off the cycle transiently, but the dynamics pull it back.

A qualitatively richer situation arises when the phase-space landscape contains not one valley but several, as illustrated in Figure 1. A system is *multistable* when two or more distinct attractors coexist for the same parameter values (Feudel, 2008). Figure 1 shows the simplest example, the potential $V(x) = (x^2 - 1)^2$: two stable states at $x = \pm 1$ are separated by an unstable fixed point at $x = 0$. Each attractor has a basin of attraction; the boundary between basins is the *separatrix*. Two runs of the same experiment with identical parameters but starting on opposite sides of the separatrix converge to entirely different long-run states. When many runs of the same system are observed, the distribution over which attractor each run reached reflects the geometry of the basins. The double-well of Figure 1 is a deterministic, continuous potential, whereas the debate we study is a finite-state stochastic chain. In that setting the analogue of coexisting attractors is a set of competing *absorbing states*, and the analogue of basin volume is not a geometric area but a probability measured relative to the initialisation distribution, since it is that distribution which decides how often each absorbing state is reached (Menck et al., 2013).

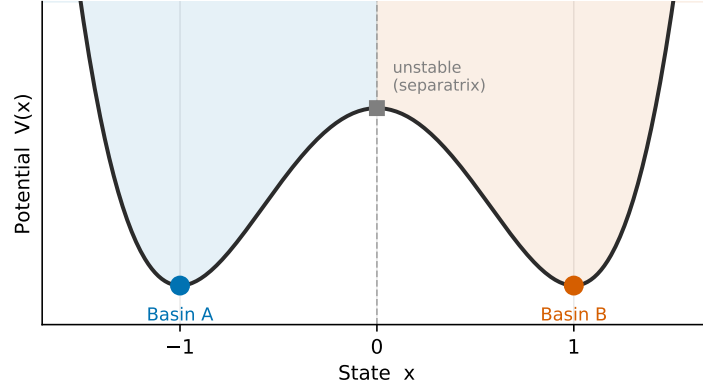


Figure 1: The potential $V(x) = (x^2 - 1)^2$ has two stable attractors at $x = \pm 1$, separated by an unstable fixed point at $x = 0$. The shaded regions are the two basins of attraction. Initial conditions in Basin A converge to the left attractor; those in Basin B converge to the right attractor. Which long-run state the system reaches depends on its initial condition and any fluctuations large enough to cross the separatrix.

2.2 Motifs and Opinion Dynamics

The patterns that arise in dynamical systems are not arbitrary. Across biological networks, electronic circuits, social groups, and neural dynamics, the same recurring organisational building blocks appear with consistent functional properties. Complex-systems research calls these recurring patterns *motifs*. The concept was introduced for networks by Milo et al. (2002), who defined a *structural motif* as a small subgraph that appears significantly more often in a real network than in a randomly rewired surrogate with the same degree sequence. Alon (2007) showed that the *coherent feed-forward loop*, a three-node pattern in which A influences both B and C while B also influences C, acts as a sign-sensitive delay filter under AND logic, and this functional property is conserved a property that recurs across transcription, signalling and neuronal networks. A complementary concept is the *dynamical motif* (Driscoll et al., 2024): a recurring pattern in the *trajectory* of a system rather than in its wiring. Where a structural motif is a statement about who interacts with whom, a dynamical motif is a statement about how the system behaves over time. The two are related, since structural motifs are studied precisely because each generates a characteristic dynamical regime, but they are not equivalent: the same structural motif can produce different dynamics under different parameters, and the same dynamical pattern can arise from different structural configurations.

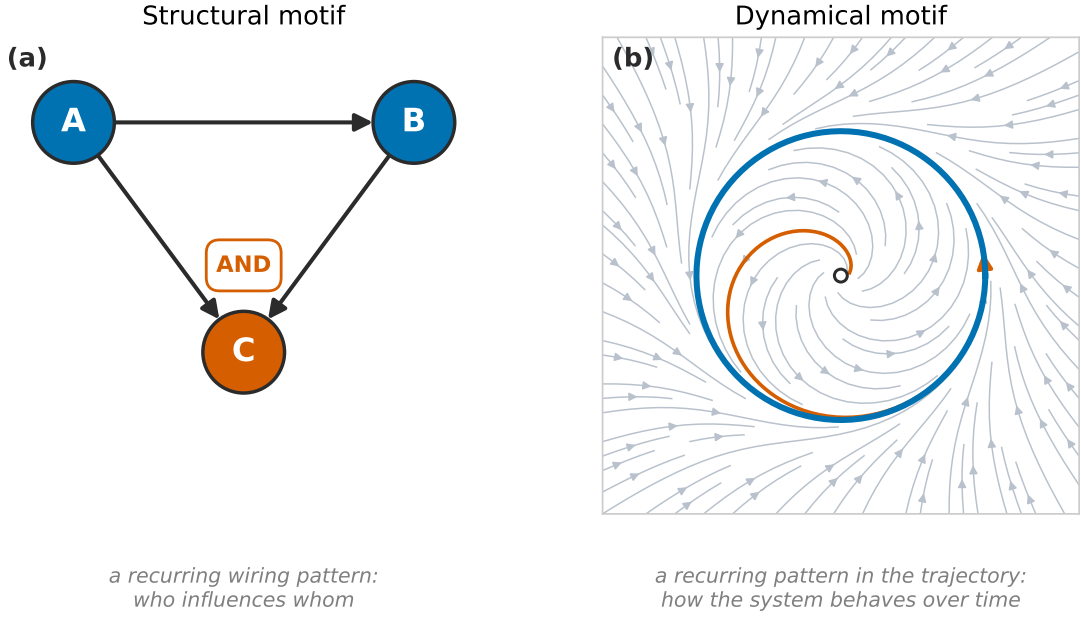


Figure 2: The two senses of *motif*. *Left*: a structural motif is a recurring wiring pattern; the coherent feed-forward loop shown here has A influence both B and C while B also influences C, so under AND logic C changes only once both of its inputs do. It is a statement about who influences whom. *Right*: a dynamical motif is a recurring pattern in the trajectory; here a trajectory (grey) spirals onto a limit cycle (blue), a self-sustained periodic orbit. It is a statement about how the system behaves over time. The same wiring can yield different dynamics, and the same dynamics can arise from different wiring.

Positive feedback is a mechanism that links structural and dynamical motifs: deviations are amplified rather than corrected, turning an early advantage into a self-reinforcing lead. Arthur (1989) gave the canonical social example with technology lock-in, where increasing returns keep an early leader ahead even when alternatives are objectively superior. Positive feedback can also operate through network structure: a node positioned to receive and relay signals from many others accumulates causal weight, concentrating influence through its position rather than through any informational advantage. Understanding whether such concentration is a property of the prescribed topology or of the dynamics themselves requires distinguishing *structural connectivity* from *effective connectivity* (Friston, 2011). Structural connectivity is the static graph encoding who is allowed to communicate with whom. Effective connectivity is the causal influence one unit actually exerts on another across a recorded trajectory; it can be estimated from the trajectory under a model of how influence operates, even when the prescribed wiring is fixed. The same structural graph can give rise to very different effective-connectivity patterns depending on which nodes happen to speak with authority at critical moments.

For the specific setting of agents forming and updating beliefs through repeated local interaction, there is a dedicated family of formal models: *opinion dynamics*. The models differ in what they assume about the opinion variable (continuous or discrete), the update rule (deterministic or stochastic), and who listens to whom (all peers, similar others, or

no one). The foundational DeGroot model (DeGroot, 1974) considers N agents, each holding a real-valued opinion $p_i(t)$. At each step, every agent replaces its opinion with a weighted average of all agents’ current opinions:

$$p_i(t+1) = \sum_{j=1}^N T_{ij} p_j(t), \quad \text{i.e., } \mathbf{p}(t) = T^t \mathbf{p}(0), \quad (2)$$

where T is a row-stochastic trust matrix ($T_{ij} \geq 0$, $\sum_j T_{ij} = 1$). The model contains no ground truth and no external information. When the trust network is strongly connected and aperiodic, the process converges to a consensus $c = \boldsymbol{\pi}^\top \mathbf{p}(0)$, where $\boldsymbol{\pi}$ is the unique left eigenvector of T with eigenvalue 1, normalised so that $\sum_i \pi_i = 1$. The vector $\boldsymbol{\pi}$ is called the *social power* vector (Jia et al., 2015): π_i is the weight agent i ’s initial opinion carries in the final consensus. An agent gains social power by being trusted by other high-power agents, not by being correct. The extreme case, in which almost all social power rests on a single agent while the others carry almost none, is the *autocratic* pole of the model (Jia et al., 2015); it is the opinion-dynamics form of the dominance motif we detect. Convergence to a single consensus requires the trust network to be both strongly connected and aperiodic, and each condition can fail in a way that maps onto one of our motifs. If the network contains two or more closed groups that listen only to each other, those groups settle on different consensus values, a group-level multistability; if aperiodicity fails, the process never settles but cycles through a repeating sequence of states, a limit cycle. Multistability and limit cycles are thus the two ways the same convergence guarantee can break down.

Three refinements of this basic picture recur in what follows. Agents often hold a graded internal belief but must commit to a discrete public choice: the *continuous-opinion, discrete-action* (CODA) model (Martins, 2008) captures exactly this split and is the template for our agent state, whose latent confidence evolves continuously while its vote stays discrete (Section 4.2). When the opinions are discrete labels with no intrinsic ranking, *naming-game* and related discrete-consensus models describe how a population settles on a shared convention (Baronchelli, 2018), the regime our multistability analysis appeals to (Section 6.2.2). And a *stubborn* agent that never updates acts as a fixed anchor that can pull the group toward its position (Friedkin and Johnsen, 1999), precisely the variant our devil’s advocate probe instantiates (Section 6.3).

2.3 LLM Agents and Multi-Agent Debate

A large language model (LLM) is a neural network, in current practice a Transformer (Vaswani et al., 2017), trained to predict the next token in a sequence and aligned to follow instructions through supervised fine-tuning and reinforcement learning from human feedback (Ouyang et al., 2022). In the broad sense, “LLM agent” covers any system that places an LLM in a loop with tools, environment feedback, or other models. In the narrow sense used throughout this thesis, an *LLM agent* is an LLM in a message-passing loop with other LLMs: it receives a state comprising the task, its prior position, and messages from peers; it produces an action consisting of a written argument and a vote; and the

result is fed back as the next state. Sampling is stochastic: *temperature* controls the sharpness of the output distribution, and raising it increases response diversity at the cost of coherence, so two agents given identical prompts at a positive temperature can produce different responses. An agent’s effective state is its context, the accumulated text it conditions on, and context length is finite; beyond a certain history depth, earlier rounds must be truncated or summarised, which is why the extent of retained interaction history is a design parameter with functional consequences.

The standard single-model inference-time alternatives to multi-agent protocols are *chain-of-thought* prompting (Wei et al., 2022), which improves reasoning by eliciting step-by-step intermediate text from one model, and *self-consistency* (Wang et al., 2023), which samples several reasoning paths from a single model and returns the majority answer. Both can be applied without additional models or communication rounds and serve as the principal baselines against which multi-agent protocols are compared.

Multi-agent debate (MAD) is a protocol in which several LLM agents independently propose an answer and then exchange and revise their reasoning over multiple rounds before committing to a final decision (Du et al., 2024). Each round typically consists of agents reading the current arguments of their peers and producing an updated argument and vote in response. The group commits when it reaches sufficient agreement or a round limit is hit. The motivating argument is error correction: if agents make partially independent mistakes, deliberation lets the group surface and correct them. This gain assumes the mistakes are independent, however: correlated votes, which social influence itself induces, erode it, and beyond a threshold the majority no longer outperforms an individual (Ladha, 1992). Reported accuracy gains on factual and mathematical reasoning (Du et al., 2024, Liang et al., 2024, Chen, Saha and Bansal, 2024) and the use of debate panels as evaluators (Chan et al., 2024) have made MAD a standard building block in collaborative AI systems. The term “debate” also names a distinct proposal in AI safety where two models argue before a human judge in order to make model reasoning checkable (Irving et al., 2018); the collaborative protocol studied here descends from the reasoning-improvement line rather than from that one.

Debate designs vary along several structural axes. The *communication topology* fixes who reads whom: a fully-connected group in which every agent sees every other, versus sparser arrangements such as a star, where peripheral agents communicate only through a central hub; sparser topologies can match fully-connected debate at lower token cost (Li et al., 2024). The *memory window* fixes how much interaction history each agent retains from one round to the next, controlling the influence of earlier rounds on current decisions. The *decision protocol* fixes how individual votes are aggregated into a group answer; this choice alone can change which tasks benefit from debate (Kaesberg et al., 2025). Additionally, debate systems vary in whether agents are instances of one model or of several, and whether they are assigned distinct personas or roles, a dimension the literature explores as a means of introducing deliberate diversity into the group (Liang et al., 2024). How these design choices behave in practice, and the growing evidence that debate can also amplify errors rather than correct them, is the subject of the related work reviewed next (Section 3).

3 Related Work

This chapter locates the thesis in the literature in three passes. Section 3.1 and Section 3.2 survey what multi-agent debate does and how the interaction dynamics inside it have been studied; Section 3.3 weighs the available ways of analysing such systems against one another; and Section 3.4 states the gap this thesis addresses.

3.1 Multi-Agent Debate: Design and Critique

The multi-agent debate (MAD) paradigm was introduced by Du et al. (2024), who let several instances of a language model independently propose answers and then critique and revise them over multiple rounds before committing to a joint response. This “society of minds” setup was reported to improve factual accuracy and mathematical reasoning while reducing hallucination, treating the underlying model as a black box. The core intuition, that partially independent agents make partially independent errors which deliberation can surface and correct, has since motivated a broad family of systems, from confidence-weighted round tables across heterogeneous models (Chen, Saha and Bansal, 2024) to evaluation panels that judge text more reliably than a single evaluator (Chan et al., 2024).

Two design axes recur throughout that family and are the ones we vary. Communication topology fixes who reads whom: sparse structures can match fully-connected debate at substantially lower token cost (Li et al., 2024), and across arbitrary networks the placement of a hub node shifts performance appreciably (Regan et al., 2024). Memory fixes how much interaction history each agent retains. The two were recently crossed in a single factorial by Mehdizadeh and Hilbert (2026), who find no main effect of topology on coordination success but a strong interaction with memory: longer memory slows settling in decentralised networks and accelerates it in centralised ones, so the same parameter pushes the system in opposite directions depending on the network’s structure. Faster settling in the centralised case, they note, means locking into a fragmented plateau rather than reaching agreement. Beyond these two, the decision protocol itself matters, with voting favouring reasoning tasks and consensus favouring knowledge tasks, and adding agents helping where adding rounds can hurt (Kaesberg et al., 2025).

Against this largely optimistic backdrop, a critical literature has emerged. Zhang et al. (2025) find that MAD frequently fails to outperform chain-of-thought (Wei et al., 2022) or self-consistency (Wang et al., 2023) despite far greater compute, and argue that model heterogeneity rather than debate is what reliably helps. Choi et al. (2025) show that majority voting alone accounts for most of the attributed gains, and prove that under their formalisation debate induces a martingale over belief trajectories and so cannot improve expected correctness on its own. Estornell and Liu (2024) supply the mechanism: when agents are similar in capability or in the responses they produce, debate dynamics become static and the procedure converges to the majority opinion, so that where the majority

reflects a misconception shared through common training data, debate converges to the misconception.

Two 2026 results push this further and are the closest published statements of what we measure. Oh et al. (2026) decompose the failure into a biased static initial belief and homogenised dynamics that amplify the majority view regardless of correctness. Bertalaníč and Fortuna (2026) show that unguided homogeneous debate matched or underperformed *isolated self-correction* at several times the token cost, and that a control injecting rationales from unrelated problems performed no worse than genuine peer exchange. They name the resulting loss an *oracle gap*, the distance between the accuracy achievable from the team’s own generation pool and the accuracy the vote actually returns, a quantity we measure directly in Section 6.1.

What unifies the affirmative and the critical strands is that both evaluate debate almost exclusively by its *final outcome*. The deliberative process that produces that outcome, and whether two systems reaching the same accuracy do so through genuinely different dynamics, remains largely outside the scope of these evaluations.

3.2 Interaction Dynamics in LLM Collectives

A distinct and more recent line looks inside the debate at how positions evolve and influence one another. Each of our four motifs corresponds to a debate phenomenon that this literature documents to a very different degree, and we take them in turn, before turning to the opening vote distribution, which is not itself a motif but the input on which they all act.

Rising confidence. Prasad and Nguyen (2025) stage repeated adversarial debates and find pervasive miscalibration: models begin overconfident, grow *more* confident that they are winning as the exchange proceeds even absent new supporting evidence, and frequently both claim victory at once, a logical impossibility. That this drift is independent of correctness has since been confirmed directly, with pre-communication confidence a near-worthless predictor of whether an agent is right, and peer exchange further eroding what little discrimination confidence carries (Bertalaníč and Fortuna, 2026). Upward, anti-Bayesian and correctness-blind confidence drift is exactly the kind of positive-feedback signature that motivates a dynamical treatment.

Conformity and influence concentration. Wynn et al. (2025) show that debate can *decrease* accuracy over time even when stronger models outnumber weaker ones, as agents abandon correct answers rather than challenge flawed arguments; they examine sycophancy (Sharma et al., 2024) and social conformity among the contributing factors. Cho et al. (2025) find the gap between an agent’s own confidence and its perceived confidence in peers governs how likely it is to conform, and Mehdizadeh and Hilbert (2025b) characterise the dose-response as a sigmoid with a model-specific threshold. Cau et al. (2026) identify an aggregate directional asymmetry they call *agreement drift*, which survives in balanced populations and so cannot be reduced to a majority effect. Where influence concentrates rather than diffusing, an emergent leader-follower structure forms (Chen

et al., 2023), and outcomes come to depend heavily on the competence of whichever agent occupies the centre (Han et al., 2026), a dependency our star topology is designed to expose.

Premature lock-in. Chuang et al. (2025) report that role-playing LLM groups converge far more strongly and rapidly than human groups do, indicating a systematic bias toward premature agreement. Purely cooperative framings sharpen this: with no dissenting pressure a group can collapse into a mutually reinforcing echo chamber. A structurally sharper version is captured by HiddenBench (?), which adapts the hidden-profile paradigm from social psychology (Stasser and Titus, 1985, Schulz-Hardt and Mojzisch, 2012): when the evidence needed for the correct answer is distributed across agents, collective accuracy collapses far below what a single fully-informed agent achieves, because agents lock onto shared information and fail to pool their private knowledge. We adopt HiddenBench as our second benchmark for exactly this reason.

Oscillation and non-termination. The fourth regime, sustained oscillation, is conspicuous by its near-absence from this literature. A debate that never converges but cycles through the same positions is a limit cycle in the dynamical sense (Strogatz, 2015), yet debate studies typically impose a fixed round budget and fold any run that fails to settle into an undifferentiated “no consensus” category, so a periodic orbit is never distinguished from slow convergence. We flag this near-silence because it prefigures our own finding that genuine limit cycles are essentially absent outside the star topology (Section 6.2.4).

The opening distribution. That round-zero answers govern the outcome is stated in several forms. Hu et al. (2025) make it a formal premise, proving debate improvement over majority vote only under the assumption that at least one initial response is generated under the correct concept. Zhu et al. (2026) report that initial answer diversity correlates significantly with final accuracy and propose a diversity-aware initialisation to raise the chance that a correct hypothesis is present at the start. Empirically, Wu et al. (2025) find that an agent starting inside a correct majority is almost perfectly stable while one facing an incorrect majority recovers in only a minority of cases (3.6–34.4% depending on the model). In every case the opening distribution is observed as it arises rather than constructed, leaving its effect confounded with task difficulty and agent confidence, which is what Section 6.3 is designed to separate.

The vocabulary for naming these phenomena precisely, network and dynamical motifs, positive feedback and lock-in, opinion dynamics and social power, and the distinction between structural and effective connectivity, was introduced in Section 2.2 and is not repeated here. What that section leaves open is whether the vocabulary is more than an analogy once it is carried to LLM populations.

That it is more than an analogy is established empirically. Ashery et al. (2025) demonstrate the spontaneous emergence of universally adopted conventions in decentralised populations, show that strong collective biases arise even when individual agents exhibit none, and find that committed minorities can impose alternative conventions on the majority, the LLM analogue of the critical-mass threshold measured in human groups (Centola et al., 2018). The same behaviour can be made analytic: Okawa (2026) derives a mean-

field treatment of debate from spin models, predicting a conformity threshold above which a collective norm emerges, smoothed into a crossover at the small group sizes typical of debate.

3.3 Ways of Analysing Multi-Agent Systems

Before stating the gap, it is worth making explicit *why* a complex-systems treatment is the appropriate instrument, by situating it against the alternatives. Each lens answers a different question and carries different costs.

Outcome evaluation. The dominant approach scores a system by final task performance (Du et al., 2024, Zhang et al., 2025, Choi et al., 2025). It is objective, comparable across systems, and measures what practitioners care about. But an outcome is a scalar summary of a trajectory: two systems reaching the same accuracy, one by genuine error correction and the other by a confident slide into a shared mistake, are indistinguishable under it. It says *whether* a configuration worked, not *why*.

Mechanistic interpretability. Opening the models themselves operates at the wrong level of abstraction for our question. Consensus, oscillation, and influence concentration are properties of the *group* trajectory rather than of any single forward pass (Miehling et al., 2025), and internals are inaccessible for the API-served models typical of applied multi-agent systems.

Qualitative transcript analysis. Reading transcripts directly is rich and interpretable, and we draw on it informally when reading the mechanism behind flagged limit-cycle cases (Section 6.2.4). On its own it is subjective, does not scale to tens of thousands of repetitions, and yields no falsifiable structure testable against a null.

Influence attribution. A fourth approach quantifies directed influence between agents, either information-theoretically through transfer entropy over the utterances of interlocutors (Goodall et al., 2026), or by treating influence as a cooperative game and estimating it through agent removal (Lu et al., 2026, Cui et al., 2025). These yield causal or information-theoretic estimates rather than the adoption-based signature we use, at the price of model access or repeated re-execution; we return to the trade-off in Section 7.4.

Learned diagnostics from logs. A fifth and very recent approach trains a predictor on log-derived features, either to decide whether a query warrants debate at all (Fan et al., 2026) or to decide when a majority vote should be overturned (He et al., 2026). These establish that ordinary logs carry enough signal to support useful decisions. They are, however, predictive rather than explanatory: they learn that a feature configuration precedes failure without naming which dynamical regime the system is in, and they require labelled outcomes to train.

The dynamical lens. The approach taken here treats the debate as a dynamical system and measures recurring patterns in its state trajectory (Strogatz, 2015, Driscoll et al., 2024). Its advantages match the gaps above: it operates at the group level, scales to the

full dataset, needs only interaction logs and no labelled outcomes, and gives each motif a falsifiable signature testable against a principled null. Its disadvantages must be stated as plainly. Transferring motif concepts from physical and biological systems to an LLM collective is a modelling choice, not a law; the detection thresholds are partly conventional; a four-agent group is small for some of these measures; and, absent intervention, the detectors identify *signatures* and *associations* rather than certified attractors or causes. We adopt this lens because it is the only one that can name and measure the pathologies documented above while remaining applicable to black-box models, but we carry its limitations into the interpretation of every result (Section 7.4).

3.4 Research Gap

The literature surveyed above has converged on a diagnosis without converging on an instrument. Outcome-centric evaluation, from the proponents of debate and its critics alike, measures where a deliberation ends and leaves how it got there unexamined. The dynamics literature does examine the trajectory, but documents one pathology at a time, overconfidence here, herding there, premature consensus elsewhere, and relates them to one another through a shared cause such as sycophancy rather than through the formal dynamical regimes they resemble. None of these signatures is tested against a null that says what the same statistic would look like by chance, so a raw detection rate cannot be distinguished from the ordinary regularity of a small, discrete state space.

More consequentially, where the complex-systems vocabulary *has* been applied to LLM collectives, it has been applied on substrates from which truth was deliberately removed. Ashery et al. (2025) study arbitrary conventions where any label is an acceptable equilibrium. Mehdizadeh and Hilbert (2026) verify that their convention symbols carry no intrinsic model preference, precisely so that dynamics are not confounded by correctness. Cau et al. (2026) choose a non-empirical proposition explicitly to avoid convergence driven by factual correctness. This is a principled choice, and it is what makes those studies clean: if the goal is to understand how a group settles on *a* convention, having no better convention available is exactly right. It also leaves them silent on the question a practitioner faces, which is not whether the group locks in but whether it locks in on the right answer—the epistemic case, where some claims are better supported than others, remains open.

This thesis takes up that question. The gap it addresses is the absence of an empirical, literature-grounded detection of self-organization motifs in *truth-valued* deliberation, and our response follows the three research questions of Section 1.2. As an *instrument*, we define detectors for self-reinforcement, multistability, dominance, and limit cycles, each grounded in the corresponding literature and each equipped with its own null model, and we run them over a repetition structure that makes across-repetition attractor statistics computable at all; prior work runs a single simulation per configuration (Cau et al., 2026) or analyses run-level means without across-run endpoint distributions (Mehdizadeh and Hilbert, 2026), and so cannot measure multistability even in principle. As a *characterisation*, we apply the suite where every dynamical quantity can be conditioned on

correctness, across two deliberately contrasting information regimes. As a test of *causation*, we construct the opening vote composition rather than observing it, separating its effect from the difficulty and confidence it co-varies with elsewhere. And throughout, we treat the *absence* of a motif as a finding rather than a failed measurement, a stance the outcome-centric literature has no natural place for.

4 System and Experimental Setup

The premise of this thesis is that a multi-agent debate is a dynamical system whose trajectory carries measurable structure. Turning that premise into empirical claims requires a system built as a *measurement instrument*, that is not exclusively focusing on a high-performing pipeline. Thus, the protocol must expose a discrete vote and a confidence score per agent and per round, with private belief cleanly separated from public message, so that the interaction log is a faithful record of the underlying dynamics. Furthermore, the two structural axes, communication topology and memory window, must be treated as *controlled perturbation axes* so that we can ask which dynamics they cause rather than only observe.

4.1 Notation and System Model

We study a fixed group of N language-model agents that repeatedly answer the same multiple-choice question over a sequence of discrete rounds. All agents are instances of a single frozen model f_θ ; they differ only in the private context each one has seen. Because the same symbols recur throughout the detection methodology of Section 5, we fix the notation here. Table 1 summarises the symbols used across this thesis.

Symbol	Meaning
N	number of agents ($N = 4$ throughout)
i, j	agent indices, $i \in \{1, \dots, N\}$
t	round index, $t \in \{0, 1, \dots, T\}$
T	hard cap on the number of rounds
r	repetition index (an independent re-run of a task-config cell)
R	number of repetitions per task-config cell
M	number of answer options a task presents, $M \in \{3, 4\}$
v_i^t	vote of agent i at round t , one of the M options
c_i^t	private confidence of agent i at round t , $c_i^t \in \{1, \dots, 10\}$
$\mathcal{N}(i)$	neighbourhood of agent i under the communication topology
W	memory window (rounds of history each agent may see)
u	early-stopping threshold (consecutive unanimous rounds)
s_t	system macrostate at round t (the vote composition, see below)

Table 1: Core notation used from this Section onward. Rounds are indexed by t and repetitions by r ; the two must not be confused.

Three units of analysis must be kept apart, since they nest inside one another and are easy to confuse. A *round* is a single synchronous time step t within one debate, at which every agent (re)casts its vote; a debate runs for up to T rounds. A *repetition* (indexed by r) is one complete debate on a given task from round 0 to termination; because sampling is stochastic, the same task run twice yields different repetitions. A *task-config cell* is a

single question paired with one configuration (a topology and a memory window), and we execute R repetitions of each cell. In short: many rounds make up one repetition, and many repetitions make up one task-config cell.

With the notation established, two structural ingredients complete the system model. The *interaction structure* is a binary adjacency matrix $\mathbf{G} \in \{0, 1\}^{N \times N}$, where $G_{ij} = 1$ means agent i receives agent j 's messages. Self-loops are excluded ($G_{ii} = 0$), so no agent is fed its own output as a neighbour signal, and the neighbourhood of agent i is $\mathcal{N}(i) = \{j \mid G_{ij} = 1\}$. Any topology can be instantiated by supplying a different matrix without changing the agent or round logic. We use two topologies:

- **Fully connected:** $G_{ij} = 1 - \delta_{ij}$, so every agent observes all others each round. This maximises information flow and serves as the baseline. With $N = 4$ agents, each round involves $N(N - 1) = 12$ messages.
- **Star:** one hub agent is connected to all $N - 1$ spokes, and the spokes are connected only to the hub. The hub is drawn uniformly at random per repetition, independently of agent indices, so that any observed hub effect is structural rather than an artefact of indexing. Each round involves $2(N - 1) = 6$ messages, half the fully-connected cost. Star therefore trades per-round token cost for less information flow: the hub is the sole information bottleneck, so agreement typically takes more rounds to propagate.

The second structural ingredient is the system-level state variable. Individual agents are exchangeable: only how many of them hold each option matters for the system-level analyses, not which agent holds which. We therefore summarise the group at round t by its *vote composition*, the count of agents on each option,

$$s_t = (n_1^t, n_2^t, \dots, n_M^t), \quad \sum_{k=1}^M n_k^t = N, \quad (3)$$

where n_k^t is the number of agents voting for option k at round t . The number of agents N and the number of options M are independent: $N = 4$ throughout, with $(4, 0, 0, 0)$, $(3, 1, 0, 0)$, $(2, 2, 0, 0)$, $(2, 1, 1, 0)$, $(1, 1, 1, 1)$ as the possible macrostates for $M = 4$ tasks. This macrostate is the object on which the system-level motif detectors of Section 5 operate.

4.2 Debate Protocol

The protocol specifies three things: the persistent state each agent carries across rounds, the within-round message passing that produces a new state, and the memory window controlling how much past history each agent may see. We follow the multi-agent debate paradigm of Du et al. (2024), in which agents independently answer and then revise over several rounds, but instrument it so that the interaction log exposes exactly the quantities the motif detectors require.

These requirements translate into a specific per-agent state. At each round t , agent i produces four fields:

- v_i^t : the discrete **vote**, one of the task’s M options;
- $c_i^t \in \{1, \dots, 10\}$: a private **confidence**, never transmitted;
- priv_i^t : private chain-of-thought **reasoning**, never transmitted;
- pub_i^t : a **public message**, broadcast to neighbours.

The separation of the public message from the private reasoning and confidence is intentional as it lets us measure whether agents say what they privately hold. Furthermore, the split of vote and confidence follows the continuous-opinion, discrete-action template of Martins (2008), and the public/private channel separation follows the DEBATE design of Chuang et al. (2025).

Building on this state structure, for rounds $t \geq 1$ each agent also emits an ephemeral *argument* before committing to a belief:

$$\text{argument}_i^t = (\text{defense}_i^t, \text{challenge}_i^t, \text{question}_i^t), \quad (4)$$

consisting of a **defense** (one concrete reason the current vote is correct), a **challenge** (a dispute of a specific peer claim, or a concession), and a **question** (a targeted query to a named peer). The argument is visible to neighbours at the start of the *Update* phase but does not itself change any vote. These fields structure the deliberation but are not themselves used quantitatively, but helped constructing a system of higher debate intensity.

Together, the committed state and the ephemeral argument define the round structure. Each round consists of two sequential phases; within each phase all agents act in parallel, but the *Argue* phase runs to completion before the *Update* phase begins. Round 0 is asymmetric, as no communication has yet occurred. Figure 3 illustrates the full message flow for two agents over the first three rounds.

- **Round 0 (initialisation, *Update* phase only).** Each agent receives the question and options and produces an initial belief with no peer input:

$$(v_i^0, c_i^0, \text{priv}_i^0, \text{pub}_i^0) = f_\theta(\text{question}, \text{options}).$$

- **Round $t \geq 1$, *Argue* phase.** Agent i reads its self and peer history windows $\mathcal{H}_i^{\text{self}}(t, W)$ and $\mathcal{H}_i^{\text{peer}}(t, W)$ (defined formally below) and emits an argument. Votes are *not* updated in this phase:

$$\text{argument}_i^t = f_\theta(q, \mathcal{H}_i^{\text{self}}(t, W), \mathcal{H}_i^{\text{peer}}(t, W)).$$

- **Round $t \geq 1$, *Update* phase.** Agent i additionally reads its own and its neighbours' current-round arguments, then updates its full state; if it changes its vote it is instructed to cite the specific argument that caused the change:

$$(v_i^t, c_i^t, \text{priv}_i^t, \text{pub}_i^t) = f_\theta(q, \mathcal{H}_i^{\text{self}}(t, W), \mathcal{H}_i^{\text{peer}}(t, W), \{\text{argument}_k^t\}_{k \in \{i\} \cup \mathcal{N}(i)}).$$

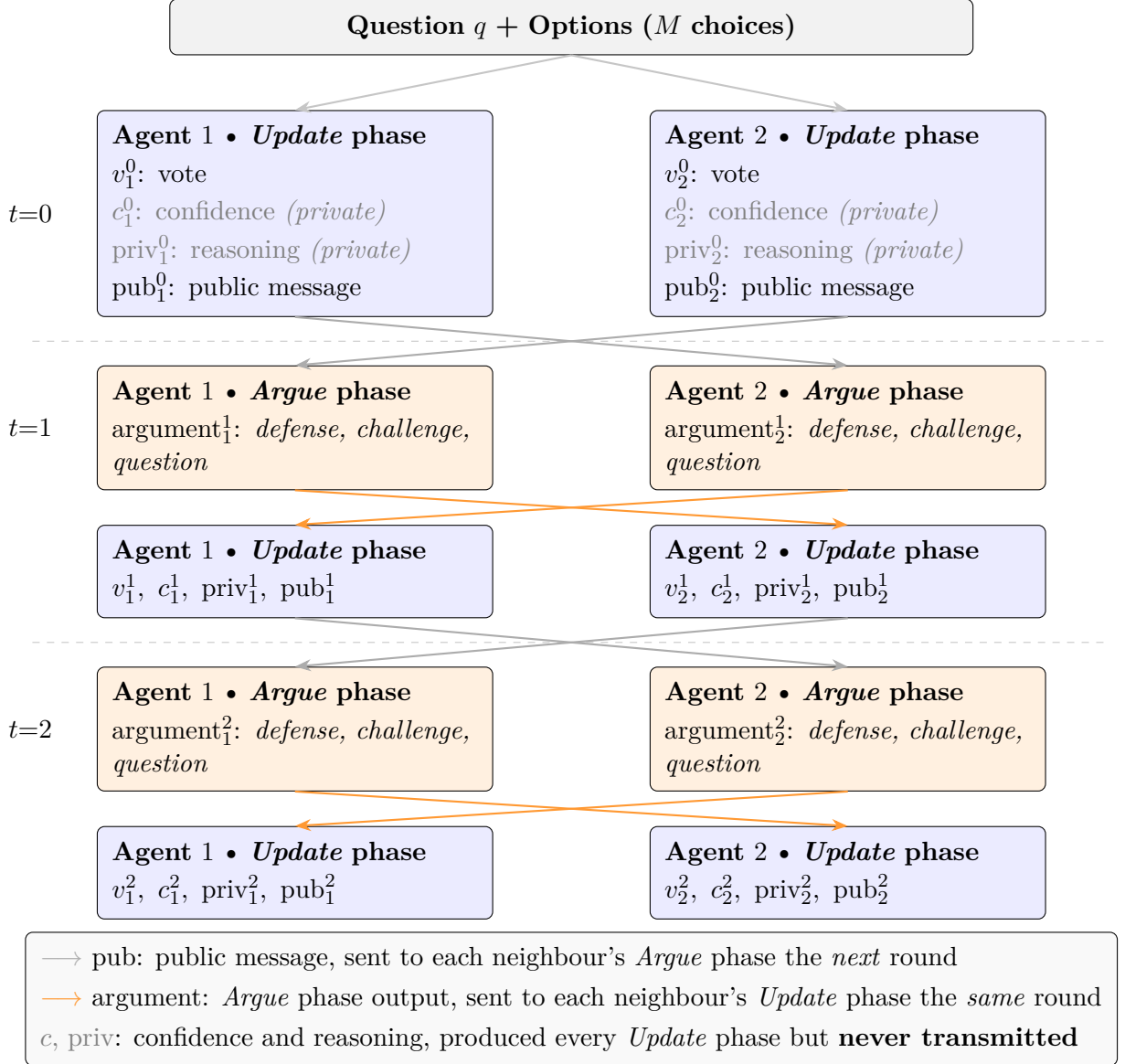


Figure 3: Two-phase synchronous protocol illustrated for two agents over rounds $t=0, 1, 2$. As round 0 has no *Argue* phase, the agents read the question directly. Each *Update* phase commits the four state fields ($v, c, \text{priv}, \text{pub}$); the confidence c and reasoning priv stay private (grey). A public message reaches neighbours in the *next* round's *Argue* phase; an *Argue* argument reaches neighbours in the *same* round's *Update* phase, which is what lets one round carry a full question–answer exchange.

The two-phase structure compresses a question–answer exchange into a single round as

a one-phase synchronous protocol could not achieve this without a two-round lag. For the dynamical analysis, only the committed *Update* state is treated as the system state at round t ; the arguments mainly function as message memory and debate facilitator. All outputs are returned as structured JSON and no problems with malformed responses were observed. The complete prompt templates are shown in Section A.3.

One aspect of the round structure left unspecified so far is how much history each agent may access. The memory window $W \in \mathbb{Z}_{\geq 1}$ controls how many past rounds each agent sees. At round t , agent i has a *self* window of its own recent states and arguments, and a *peer* window of its neighbours' recent votes, public messages, and arguments:

$$\mathcal{H}_i^{\text{self}}(t, W) = \left\{ \left(v_i^{t-k}, c_i^{t-k}, \text{pub}_i^{t-k}, \text{argument}_i^{t-k} \right) \right\}_{k=1}^{\min(t, W)}, \quad (5)$$

$$\mathcal{H}_i^{\text{peer}}(t, W) = \left\{ \left(v_j^{t-k}, \text{pub}_j^{t-k}, \text{argument}_j^{t-k} \right) \right\}_{j \in \mathcal{N}(i), k=1 \dots \min(t, W)}. \quad (6)$$

Peer entries are presented in randomised order at each call to mitigate position bias, and round-0 entries omit the argument field, which does not exist yet. A larger memory window W exposes more history but also interacts with documented long-context degradation (Liu et al., 2024), which is why we treat W as an explicit perturbation axis rather than fixing it implicitly. $W = 1$ is the near-Markov baseline in which each agent sees only the previous round.

Now that the protocol and memory structure are defined, the remaining question is when to stop. Running every repetition for the full T rounds is wasteful once agents have converged: pilot transition matrices show a self-loop probability of 0.97–1.00 for the unanimous states, meaning that once consensus is reached, escape is rare. We therefore stop a repetition as soon as u consecutive rounds are unanimously agreed. The threshold u trades efficiency against reliability: too small and a transient consensus may be cut short; too large and the savings shrink. Analysing pilot repetitions (the efficiency trade-off is detailed in Appendix A.2), we find that the harmful case, in which stopping would have prevented a later recovery to the correct answer, stays negligible while coverage remains high, and we set $u = 3$ with a hard cap of $T = 15$. Terminating debate at consensus is standard practice in the MAD literature (Yin et al., 2023, Yang et al., 2025, Kaesberg et al., 2025).

When a repetition ends, the system's answer is the *plurality* of the agents' votes at the final round, the option held by the most agents. Because a repetition stops only after $u = 3$ consecutive unanimous rounds, every converged repetition ends with all N agents on one option, so the plurality is unambiguous and equals that option. A tie can therefore arise only in the rare repetitions that reach the hard cap $T = 15$ without converging, which occurs in approximately 3.8% of repetitions. Such ties are broken deterministically towards the first option in canonical order. The resulting label is a low-confidence artefact of wherever the unresolved run happened to land rather than a genuine group decision, and Section 6.2.4 recommends flagging these repetitions as unresolved instead of trusting the reported majority.

4.3 Datasets

Because this work studies motif emergence rather than task-completion performance, our benchmark selection criteria differ from those usual in the MAD literature. A benchmark suitable for motif detection must provide (i) a **discrete shared decision variable**, so that all agents commit to a position in the same finite set each round; (ii) **verifiable ground truth**, so dynamics can be related to correctness; (iii) **calibrated difficulty**, since saturated tasks remove the disagreement that drives deliberation; and (iv) **sufficient initial disagreement**, without which no motif can be distinguished from trivial consensus.

Applying these criteria rules out most existing benchmarks. Saturated reasoning sets such as GSM8K and MMLU leave no headroom for deliberation dynamics; native multi-agent benchmarks are largely incompatible, being either role-specialised or defined over spatial rather than categorical positions. A full survey of the candidates we considered and the reason each was rejected is given in Section A.1. Two benchmarks satisfy all four criteria and are complementary, so we use them throughout this study.

GPQA-Diamond. GPQA-Diamond (Rein et al., 2024) is a graduate-level science question-answering benchmark of 198 questions in biology, physics, and chemistry, written by domain experts and validated through a “Google-proof” protocol that resists retrieval-based shortcuts. It satisfies all four criteria directly: four-option multiple choice ($M = 4$) provides the discrete decision variable; expert validation gives unambiguous ground truth; at release, frontier models were well below ceiling (Rein et al., 2024), and while the most capable models have since reached $\sim 85\%$ (Tian et al., 2025), mid-tier models remain well below, leaving genuine headroom for debate. GPQA-Diamond also has direct multi-agent precedent (Tian et al., 2025, Kenton et al., 2024, Wang et al., 2025). We adopt it as the **primary** dataset. The main caveat is that this benchmark has less cumulative MAD literature compared to MMLU or GSM8K, which is however mostly the case as it is a more recent benchmark and not because of a methodological pivot. A second caveat concerns comparability of the accuracy numbers: published figures refer to strong models on the full benchmark, whereas we use a mid-tier model on a screened task subset only. Our results are therefore not comparable to published frontier GPQA-Diamond numbers, and are not intended to be; the screening pass deliberately selects the most dynamically contested tasks, where accuracy is low by design.

HiddenBench. Moreover, we use HiddenBench (?) as a structurally complementary **secondary** dataset. It formalises the hidden-profile paradigm from social psychology (Stasser and Titus, 1985, Schulz-Hardt and Mojzisch, 2012), in which the evidence needed for the correct answer is split across agents, so that no single agent can solve the task alone and success requires information pooling. Its value here is as a control for a specific confound. Motifs seen only on GPQA-Diamond might be artefacts of cooperative reasoning over a reliable individual signal, in which RLHF-induced truth-seeking (Ouyang et al., 2022) amplifies convergence. Under the hidden-profile condition, by contrast, pre-discussion accuracy is very low by construction (?), so any post-discussion improvement must come from social information transfer rather than from individual competence. Mo-

tifs that appear on both datasets are therefore robust to this confound, while motifs that appear on only one are themselves informative on a dataset-level.

One structural detail of HiddenBench matters for the analysis. While we use $N = 4$ agents throughout, the number of answer options is task-dependent, $M \in \{3, 4\}$. The agent count N and the option count M are therefore independent quantities and should not be conflated. This is why every option-indexed quantity in Section 5 is defined over the task’s actual M options. Extension to mixed-motive negotiation datasets is left to future work.

4.4 Experimental Design

The system exposes many tunable components. We classify each as **scan** (a primary perturbation axis), **fix** (held constant), or **defer** (out of scope). Table 2 summarises the classification and the literature supporting it; the detailed per-knob justification is deferred to Section A.2.

Of these components, two are varied as primary perturbation axes, chosen because they are the canonical levers for network-motif analysis:

- **Memory window** $W \in \{1, 2, 5\}$, with $W = 1$ the near-Markov baseline. Memory content is an established driver of debate behaviour (Du et al., 2024, Mehdizadeh and Hilbert, 2026, Zargari Marandi, 2026).
- **Topology** $\in \{\text{FC}, \text{STAR}\}$. Fully-connected probes the symmetric baseline; the star probes hub-induced dominance, following evidence that centralised structures concentrate influence (Li et al., 2024, Chen et al., 2023, Regan et al., 2024).

Their product gives $3 \times 2 = 6$ configurations per task. The dataset (GPQA or HiddenBench) is classified as a scanned condition in Table 2 because the information condition varies between benchmarks, but it is not a structural axis in the sense of a manipulation applied within a fixed task; the two benchmarks are instead analysed separately. The full design therefore spans $6 \times 2 = 12$ configuration–dataset cells. All remaining components are held constant to isolate the effect of the two scan axes:

- **Agents** $N = 4$: the smallest value at which the two topologies are structurally distinct and multi-way splits can occur.
- **Rounds** $T = 15$ (hard cap) with early stopping $u = 3$: T is an observation budget rather than a structural knob, since trajectories can be cropped post hoc but not extended.
- **Model** `mistral-medium-instruct`: a single mid-tier model to isolate topology and memory effects from capability-asymmetry effects of heterogeneous models (Wynn et al., 2025).

- **Sampling** temperature 0.7, reasoning disabled: diverse yet focused outputs; for homogeneous agents stochasticity arises solely from sampling (Zhu et al., 2026), so a positive temperature is necessary for any response diversity.
- **Framing** cooperative, with the full *Argue/Update* structured output as the measurement instrument, and synchronous execution to remove agent-order confounds.

Finally, selecting which tasks to run completes the design. Motifs are per-task properties, and running the full grid at high R on every question is infeasible: at roughly 25 s per repetition, the full GPQA-Diamond set across six configurations would take approximately 10.3 days (248 h) at $R = 30$ or 17.2 days (413 h) at the final $R = 50$; a two-phase design that employs an initial screening process reduces this to approximately 29 h. It begins with a shallow *screening* pass which runs every task once at a single baseline configuration (FC, $W = 1$, $R = 3$) and retains only those tasks where debate actually occurs *and* the final answer varies across repetitions, since only those tasks have dynamics, rather than the model prior, deciding the outcome. Tasks that are unanimous at $t = 0$, that always converge to the same answer, or where no genuine disagreement arises, are excluded.

Knob	Verdict	Value	Key references
Memory W	scan	$\{1, 2, 5\}$	(Du et al., 2024, Mehdizadeh and Hilbert, 2026, Zargari Marandi, 2026)
Topology	scan	fc, star	(Li et al., 2024, Chen et al., 2023, Regan et al., 2024)
Agents N	fix	4	(Du et al., 2024, ?)
Rounds T	fix	15 (cap), $u = 3$	(Kaesberg et al., 2025, Yang et al., 2025)
Output structure	fix	full <i>Argue/Update</i>	(Liang et al., 2024, He and Feng, 2025)
Framing	fix	cooperative	(Liang et al., 2024, Du et al., 2024)
Synchrony	fix	parallel-synchronous	(Du et al., 2024, Chan et al., 2024)
Model	fix	mistral-medium-instruct	(Zhang et al., 2025, Wynn et al., 2025)
Temperature	fix	0.7, no reasoning	(Zhu et al., 2026)
Information cond.	scan (dataset)	GPQA ($M=4$), Hidden-Bench ($M \in \{3, 4\}$)	(?Stasser and Titus, 1985)
Tool access	defer	none	(Rein et al., 2024)

Table 2: Classification of tunable components. Full justification per knob is given in Section A.2.

Retained tasks are promoted to a deep *characterisation* pass on the full six-configuration grid. This yields the final dataset used throughout Section 6: **50 tasks for each of the two benchmarks, six configurations each, $R = 50$ repetitions**, for a total of

$50 \times 2 \times 6 \times 50 = 30,000$ repetitions. GPQA-Diamond yields more eligible tasks than needed in the screening pass, so we retain 50 to match the number of eligible HiddenBench tasks, giving a balanced design with equal task counts per benchmark; the 50 are drawn uniformly at random from the eligible GPQA tasks, so that no further selection is made on any outcome-derived quantity. Because $R = 3$ in the screening pass introduces some noise, assignments near the inclusion boundary are mildly unstable, which we accept as a screening cost rather than a final claim.

One consequence of this design should be made explicit for the interpretation of RQ2. Because the characterisation pass runs only tasks selected by the screening procedure, raw task difficulty is held roughly constant by construction: it is a *selection criterion* for the study rather than a factor we vary within it. The “difficulty regime” contrast that RQ2 refers to is therefore operationalised at the benchmark level, GPQA (a refinement regime, agents mostly start correct) versus HiddenBench (a correction regime, agents start systematically wrong). A further interpretive consequence concerns multistability. Because tasks are retained precisely because their final answer varies across repetitions under the screening configuration, the prevalence of coexisting attractors reported in Section 6.2.2 is conditioned on this selection and should not be read as an estimate of motif frequency in an unselected task population; see Section 7.4 for discussion.

Code, prompts, and run scripts are available at <https://github.com/echtermeyer/LLM-Based-MAS>.

5 Detection Methodology

Each of the four motifs defined in this section is a hypothesis about the underlying debate dynamics. If the system self-organizes in the way a complex-systems reading predicts, then confidence should reinforce within a held position, repeated repetitions should settle into a small number of initial-condition-selected attractors, influence should concentrate on particular agents, and in some configurations vote sequences may oscillate persistently without converging. This section turns each hypothesis into a *detector*: a statistic computed from the interaction log together with a principled null model against which it is tested, so that a positive result reflects genuine structure rather than the chance regularities of a small, discrete state space. Together the four detectors form SOMA, the detector suite this thesis contributes.

Two properties are shared by all four detectors and therefore are worth stating up front. Firstly, each detector reads only from ordinary interaction logs, the vote and confidence channels defined in Section 4, which is what later lets us propose the same detectors as diagnostic instruments. And secondly, each detects a *signature*, not a certified dynamical object. Thus, we add a “Limitations” paragraph to each detector which makes explicit what the corresponding statistic can and cannot claim.

5.1 Shared Measurement Conventions

Before defining each detector individually, it is worth fixing what they share. Four things recur across every subsection below: the shape those subsections take, the classification of the motifs themselves, the macrostate the system-level detectors read from, and the nesting of the data that constrains every comparison.

The four subsections each make the same five moves, in the same order. *Concept* states the motif in dynamical-systems terms and names the quantity in a debate that plays the role of the state variable. *Measurement* defines the *measurement primitive*, the lowest-level object the detector scores, and the statistic computed from it. *Detection* gives the null model that statistic is tested against and the rule that flags the motif. *Worked example* runs the detector by hand on a small case, and *Limitations* states what the resulting statistic cannot claim. The division between the second and third move is the one worth holding on to, since it recurs throughout: measurement is what we compute from the log, detection is what we compare it against. So that the four worked examples can be read against one another, they share a single running scenario, one task with $N = 4$ agents and $M = 4$ options labelled A to D of which A is correct, and they write an agent’s state as its option with its confidence as a subscript, so that D_8 is a vote for D held at confidence 8.

The motifs those subsections detect are preselected rather than exhaustively enumerated, each motivated by a documented failure mode of LLM-based debate. Table 3 lists them together with their classification and the section in which each detector is defined. Further

motifs that fall within the same structural-dynamical framework but lie outside the scope of this study are catalogued in Appendix A.10.

Motif	Classification	Level	Section
Self-reinforcement	Structural (positive autoregulation)	Agent	Section 5.2
Multistability	Dynamical (coexisting attractors)	System	Section 5.3
Dominance	Structural (emergent hub)	System	Section 5.4
Limit cycle	Dynamical (periodic orbit)	Agent & System	Section 5.5

Table 3: The four motifs classified by type (structural or dynamical, following Section 2.2) and level (whether the detector reads from individual agent trajectories, the collective macrostate, or both).

What the table calls the *level* of a motif decides where its detector reads from. Self-reinforcement, the only agent-level motif, reads directly from per-agent vote and confidence trajectories. The three system-level readings instead operate on the agent-anonymised macrostate s_t defined in Section 4.1, whose key property is that composition retains more information than a plain vote count: a 3–1 split and a 2–2 split are distinct macrostates, and options remain labelled because one is correct and they are not interchangeable. Each of the three takes a different slice of that record, however. Multistability reads only the locked-in endpoint, dominance reads the emergent influence graph, which is built relationally and therefore needs within-repetition agent identity, and the system-level limit-cycle test reads the full sequence $s_0, \dots, s_T$.

Whichever slice a detector takes, it inherits the same nesting of the data,

$$\text{agent} \subset \text{repetition} \subset \text{task} \subset \text{config} (W \times \text{topology}) \subset \text{dataset}, \quad (7)$$

with $W \in \{1, 2, 5\}$, $\text{topology} \in \{\text{fc}, \text{star}\}$, $N = 4$, and $\text{dataset} \in \{\text{GPQA}, \text{HiddenBench}\}$, and that nesting constrains what may be compared with what. Because agents interact within a repetition, their trajectories are not independent, so the repetition is the independent unit of replication and agent-level trajectories are never pooled as if they were separate observations (pseudoreplication). For the same reason GPQA and HiddenBench are always analysed separately and never pooled, and memory-window comparisons are made within a fixed topology and dataset, since cells from different configs are not exchangeable.

A last property is shared by all four detectors, and it bounds every claim they make: each is phenomenological rather than causal. We detect a trajectory or influence *signature*, not a certified attractor, cycle, or causal hub, and separating the signature from its underlying cause requires dedicated perturbation experiments, which we leave to future work. Under the cooperative, convergence-forced design we also expect several detectors to return sparse or near-null results, which is why each detection paragraph below names the baseline its statistic is compared against rather than testing it against zero.

5.2 Self-Reinforcement

Concept. In complex systems, self-reinforcement is *positive feedback*: the current level of a state variable raises the drive toward more of the same, so deviations are amplified rather than damped (Arthur, 1989). At the motif level this is positive autoregulation, a node that augments its own production (Milo et al., 2002, Alon, 2007). For us the state variable is an agent’s expressed *confidence*. Self-reinforcement is then the tendency of an agent to push its own confidence upward while it holds a position, driven by accumulating commitment rather than by what the incoming information warrants. The pattern is already documented: in multi-turn LLM debate, confidence tends to rise across rounds in an anti-Bayesian fashion, even where no new information justifies the rise (Prasad and Nguyen, 2025).

For the detection of self-reinforcement a rising confidence trajectory is on its own necessary but not sufficient, since rational updating and peer conformity also push confidence up. We therefore measure two signatures that go beyond it. *Directional drift* is the tendency of confidence to increase rather than decrease while a position is held. *Strength scaling* is the prediction that this drift grows with how strongly an agent is re-exposed to its own prior commitments, which the memory window W controls directly. A third signature, *endogeneity*, would establish that the rise comes from the agent’s own commitment rather than from new information, but it cannot be separated from peer effects using trajectory data alone.

Measurement. The primitive is the *vote-stable segment*: a maximal block of consecutive rounds over which agent i ’s vote does not change, running from its start t_s to its end t_e , so that $v_i^t = v_i^{t_s}$ for all $t \in \{t_s, \dots, t_e\}$. Writing $L = t_e - t_s + 1$ for its length in rounds, a segment is retained only if $L \geq 3$. Two rounds would already define a slope, but that slope would be a single confidence difference with no residual degrees of freedom, so its sign would turn on one integer step; three rounds give two steps for the least-squares fit to average over. An agent might contribute several segments if it flips and one long segment if it locks in early. Agents are exchangeable, so segments are pooled within a config and never matched across repetitions. Within a retained segment we regress confidence on the round index by ordinary least squares,

$$c_i^t = \beta_0 + \beta_{\text{seg}} t + \varepsilon_t, \quad t \in \{t_s, \dots, t_e\}, \quad (8)$$

and keep the slope β_{seg} , in confidence points per round. Its sign feeds the prevalence test below, and its signed value feeds the mean slope $\bar{\beta}$ used in the W comparison.

Detection. With the segments in hand, the question is whether their slopes lean upward more often than chance allows. We read them in two ways, both conditioned on the config. The first is a presence test for directional drift. Writing n_{seg} for the number of retained segments in a config-task pair, the prevalence of upward segments is

$$\hat{p}_{\text{SR}} = \frac{|\{\text{seg} : \beta_{\text{seg}} > 0\}|}{n_{\text{seg}}}, \quad (9)$$

so flat segments ($\beta_{\text{seg}} = 0$) sit in the denominator but never in the numerator. Under a no-feedback “wobble” null an upward and a downward segment are equally likely, so with a fraction f of flat segments $P(\beta_{\text{seg}} > 0) = \frac{1}{2}(1 - f) \leq 0.5$. Thus, self-reinforcement is indicated when $\hat{p}_{\text{SR}} > 0.5$. Both the flat segments and the confidence ceiling, since a segment already at the top of the scale cannot drift further up, make a positive result conservative. The second reading uses the memory window to discriminate between mechanisms. Letting $\bar{\beta}$ be the mean segment slope in a config, an *endogenous* reading predicts that re-reading one’s own prior commitments across a longer window amplifies the internal drive,

$$\bar{\beta}(W=5) > \bar{\beta}(W=2) > \bar{\beta}(W=1), \quad (10)$$

whereas under a *peer-conformity* reading a longer window mainly adds peer content, and the ordering may be absent or reversed.

Both readings are reported per config-task and then summarised across tasks, for example “above 0.5 in k of 50 tasks”, so that an effect is shown to be broad rather than driven by one task.

Worked example. In the running scenario, suppose agent 0 holds option A for five rounds with confidences $c = (5, 6, 6, 8, 9)$. This is one vote-stable segment of length $L = 5$, and regressing confidence on the round index gives $\beta_{\text{seg}} = +1.0$ confidence points per round: an upward segment, counting toward $\hat{p}_{\text{SR}}$. Had the agent instead drifted $c = (7, 7, 6, 6, 5)$, the slope would be -0.5 and the segment would count against it. A segment that never moved, $c = (6, 6, 6, 6, 6)$, has $\beta_{\text{seg}} = 0$: retained in the denominator but not counted as upward, which is what makes the 0.5 reference conservative. Under the strength-scaling reading, the average of such slopes should grow as the window widens from $W=1$ to $W=5$.

Limitations. Beyond the shared phenomenological caveat, two points bound the claim. *Endogeneity is not isolated*: an upward segment is equally consistent with peer copying and with a peer surfacing genuinely new information in a later round, and rising confidence alone cannot tell these apart. The W ordering narrows the ambiguity but does not settle it, since a longer window also re-exposes more peer content. *Convergence is a confound*: on easier tasks fast agreement can itself raise confidence, which is why prevalence is reported per task and never as a single pooled figure.

5.3 Multistability

Concept. In dynamical systems, *multistability* is the coexistence of two or more stable attractors for one fixed parameter set. Into which attractor the system settles into is decided by the basin of attraction containing the initial condition (Feudel, 2008, Pisarchik and Feudel, 2014). At the motif level this is the dynamical signature of positive feedback with a symmetry: the same interaction rules admit several self-consistent outcomes (Milo et al., 2002, Alon, 2007). For us the state variable is the macrostate s_t , and multistability is read off the *locked-in endpoint*: for a fixed task, the debate converges to different consensus answers on different repetitions, and which one is reached is selected by the

initial vote configuration rather than by the task’s correct option. This is the structural reading of unreliable correction: on a task where the model carries a systematic bias, a wrong-consensus and a right-consensus attractor can coexist, and the group’s fate is basin competition decided at round 0, which self-reinforcement then amplifies and locks in. For the motif detection, seeing more than one final answer is necessary but not sufficient, as a task may simply be noisy. Thus, genuine multistability additionally requires that the initial condition *predicts* the endpoint (Pisarchik and Feudel, 2014).

Measurement. The primitive and independent unit is the repetition, characterised by its initial vote composition $s_0 = (n_1^0, \dots, n_M^0)$ and its final macrostate. A repetition is converged if its final macrostate is unanimous, regardless of whether that unanimity was reached via early stopping or at the round cap T . We restrict the multistability analysis to converged repetitions because a stable non-unanimous endpoint reflects a failure to reach collective agreement rather than a genuine attractor. The endpoint attractor $A^{(r)}$ of a converged repetition is the winning option, so there are at most M distinct attractors per task. Repetitions that end in a non-unanimous state are excluded. Scoring is per task-config cell with n_c denoting the number of converged repetitions in that cell. The cell’s *dominant attractor* is the most frequent endpoint option, and each repetition’s start is summarised by

$$g(s_0) \in \{0, 1, \dots, N\}, \quad (11)$$

the number of agents whose round-0 vote is on the dominant option. From these we compute the empirical basin distribution over endpoint attractors, $\hat{P}(A = k) = \frac{1}{n_c} \sum_r \mathbf{1}[A^{(r)} = k]$, a basin-stability estimate of relative basin volumes under the model’s own initial distribution (Menck et al., 2013). The effective number of coexisting attractors is the inverse Simpson (Hill) index (Hill, 1973),

$$N_{\text{eff}} = \frac{1}{\sum_k \hat{P}(A = k)^2} \in [1, M]. \quad (12)$$

$N_{\text{eff}} \approx 1$ is a single attractor, ≈ 2 two coexisting attractors, and so on, so N_{eff} reports *how many* attractors coexist. Because attractors are unanimous options, it is bounded by M . Since M varies across tasks, we report N_{eff} raw and compare it only within a fixed M . In practice GPQA is entirely $M = 4$; only HiddenBench mixes $M = 3$ and $M = 4$, so its N_{eff} summaries are reported separately by option count.

Detection. Coexistence alone does not establish multistability; the initial condition must also *select* the attractor (Feudel, 2008, Pisarchik and Feudel, 2014), and it is that association which carries the inferential claim. We measure it between $g(s_0)$ and the endpoint A with Cramér’s V (Cramér, 1946) on their contingency table,

$$V = \sqrt{\frac{\chi^2/n_c}{\min(R_{\text{tab}} - 1, C_{\text{tab}} - 1)}} \in [0, 1], \quad (13)$$

where χ^2 is the Pearson statistic and $R_{\text{tab}}, C_{\text{tab}}$ are the table’s numbers of rows (distinct $g(s_0)$, at most $N + 1$) and columns (distinct endpoints, at most M). $V \approx 0$ means the start is irrelevant; V near 1 means the start deterministically selects the endpoint. The

null shuffles the pairing of $g(s_0)$ and A across the n_c repetitions, preserving both marginals while destroying the start-to-endpoint link (Theiler et al., 1992); for $B = 1000$ surrogates the add-one one-sided p -value flags basin selection when $p < 0.05$.

The two readings combine into a single cell label,

$$\text{label}(c) = \begin{cases} \text{monostable} & N_{\text{eff}} < 1.5, \\ \text{multistable} & N_{\text{eff}} \geq 1.5 \wedge p < 0.05, \\ \text{stochastic (not multistable)} & N_{\text{eff}} \geq 1.5 \wedge p \geq 0.05, \end{cases} \quad (14)$$

with the degree of multistability read from N_{eff} and capped at M . The cutoff 1.5 is a reporting threshold, not a test; the inferential claim rests on p . The third case is the essential guard: coexistence without basin selection is scatter, not multistability. Because per-cell power is limited, results lean on the across-task distribution of N_{eff} and V rather than any single cell.

Worked example. In the running scenario, take a cell with $n_c = 30$ converged repetitions whose dominant attractor is option A . The coexistence reading depends only on the endpoint distribution, and the label depends on whether the start predicts it.

- **Monostable.** Endpoints $28 \times A$, $2 \times B$ give $\hat{P} = (0.933, 0.067)$ and $N_{\text{eff}} = 1/(0.933^2 + 0.067^2) \approx 1.14 < 1.5$: one valley.
- **Multistable.** Endpoints $15 \times A$, $15 \times B$ give $N_{\text{eff}} = 2.0$. If repetitions starting with more agents on A (high $g(s_0)$) end on A and those starting with few end on B , then $V \approx 1$, $p < 0.05$: two basins selected by the start.
- **Tristable.** Endpoints $10 \times A$, $10 \times B$, $10 \times C$ with the start predicting each give $N_{\text{eff}} \approx 3.0$, $p < 0.05$ (the maximal degree for $M = 3$).
- **Stochastic.** Endpoints $15 \times A$, $15 \times B$ again give $N_{\text{eff}} = 2.0$, but if $g(s_0)$ is unrelated to the endpoint then $V \approx 0$, $p \gg 0.05$: coexistence without basin selection, correctly *not* labelled multistable.

Limitations. First, classical basin stability draws initial conditions uniformly (Menck et al., 2013), whereas our starts come from the model’s own round-0 distribution, so $\hat{P}(A)$ estimates *effective* basin volume under that distribution, not geometric volume. Second, because $g(s_0)$ counts agents already on the eventual dominant option, a high V is consistent both with genuine basin selection and with ordinary majority-vote persistence; the test detects start-to-endpoint dependence but does not by itself separate these two mechanisms, so we read the association as necessary but not mechanistically decisive.

5.4 Dominance

Concept. Dominance is the concentration of directed influence onto a single node, an *emergent hub* which is a structural motif (Milo et al., 2002, Alon, 2007) read at the system

level, since it is a property of the whole influence distribution. Its formal home is opinion dynamics. In the DeGroot model (DeGroot, 1974) and its social-power extension, an agent’s influence equals its network centrality, and the group is *autocratic* when these are maximally nonuniform and *democratic* when uniform (Jia et al., 2015). Dominance is the autocratic side. For us the state variable is the distribution of influence across the four agents, and dominance is the tendency of that distribution to concentrate on one of them rather than to spread evenly. When it does, a single overconfident agent can pull the rest of the group toward its answer (Chen et al., 2023, Cho et al., 2025), degrading collective accuracy even when the majority would otherwise be correct (Wynn et al., 2025). To avoid attributing concentration that is simply an artefact of the wiring, we measure influence for each topology separately. The result is a concentration signature rather than proof of causation, since an adoption edge cannot distinguish genuine influence from coincidental agreement.

Measurement. The primitive and independent unit is the per-repetition influence graph, which requires within-repetition agent identity. Dominance is defined over agents and their conversions, so it is independent of the option count M . A *conversion* occurs when agent j adopts, at round $t+1$, a vote that a disagreeing neighbour i held at round t . The directed edge $i \rightarrow j$ accumulates the confidence-weighted conversions of j onto i ’s vote, with credit split among the neighbours already holding the target vote,

$$w_{ij} = \sum_t \frac{1}{m_j^t} \mathbf{1}[v_j^{t+1} = v_i^t \neq v_j^t] c_j^t, \quad m_j^t = |\{k \in \mathcal{N}(j) : v_k^t = v_j^{t+1}\}|, \quad (15)$$

where c_j^t is the converting agent’s confidence, $\mathcal{N}(j)$ its topological neighbourhood, and the factor $1/m_j^t$ shares credit equally among all eligible sources. Per-agent influence is the weighted out-strength $\sigma_i = \sum_j w_{ij}$. We use out-strength rather than eigenvector centrality (Bonacich, 1972), which is undefined or degenerate on non-strongly-connected and acyclic graphs (Newman, 2010) and hub-localises (Martin et al., 2014). Our graphs are tiny and sparse, making out-strength the only always-defined node score.

The statistic is the concentration of the resulting influence-share vector $p_i = \sigma_i / \sum_k \sigma_k$, measured by the normalised Herfindahl index (Hirschman, 1964),

$$D = \frac{\sum_i p_i^2 - \frac{1}{N}}{1 - \frac{1}{N}} \in [0, 1]. \quad (16)$$

$D = 0$ is democratic as the agents have equal influence, and $D = 1$ is a single pure hub. D is undefined when $\sum_k \sigma_k = 0$, that is when no conversion occurs at all, and such repetitions are excluded.

Detection. A high D in the star is uninformative on its own, since the centre is the only agent a leaf can be converted by and the wiring alone forces concentration. The null is therefore keyed to topology: it asks how concentrated D would be if, on this same wiring and these same conversion events, influence were assigned at random. We use a constrained shuffle in the spirit of Theiler’s “Algorithm 0” (Theiler et al., 1992) that keeps the converting agent, the round, the weight, and the number of eligible sources fixed, and redraws only the source *identity*, uniformly among $\mathcal{N}(j)$. In the fully-connected topology

every agent has three neighbours, so the shuffle can reassign any conversion to any of them and the resulting null is symmetric; concentration above it is emergent. However, for the star topology a leaf has exactly one neighbour, so a leaf conversion has only one possible source and the shuffle leaves it unchanged. Where a repetition is made up of such conversions, every surrogate reproduces the observed graph and the test can never flag. This makes the star null degenerate, so emergent concentration is identified only in the fully-connected topology. Raw D is still reported for both, but in the star it describes the wiring rather than the behaviour.

A repetition is concentrated when its observed D falls in the upper tail of its own surrogate distribution. With $B = 1000$ surrogates the add-one one-sided p -value is

$$p = \frac{1 + \#\{b : D^{(b)} \geq D\}}{1 + B}, \quad (17)$$

and we flag the repetition when $p < 0.05$. For the fully-connected topology we report the standardised excess $z_r = (D_r - \mu_{\text{null}})/\sigma_{\text{null}}$ and the flag rate against the 5% the threshold produces by chance, each summarised per task and then counted across tasks. As for self-reinforcement, an entrenching hub predicts that concentration strengthens as the memory window widens at fixed topology, and we separately report whether the hub's vote matches the final consensus and how often it capitulates.

Worked example. In the running scenario with the fully-connected topology, agent 0 opens at D_8 while agents 1, 2, and 3 open at A_6 , B_5 , and C_5 respectively. The vote sequence is

t	agent ₀	agent ₁	agent ₂	agent ₃
0	D_8	A_6	B_5	C_5
1	D_9	D_6	B_5	C_5
2	D_9	D_7	D_5	C_5
3	D_9	D_7	D_6	D_5

Three conversions occur. At round 1, agent 1 flips to D ; only agent 0 held D at round 0, so the full weight of agent 1's confidence goes to edge $0 \rightarrow 1$, giving $w_{01} = 6$. At round 2, agent 2 flips to D ; both agents 0 and 1 held D at round 1, so credit splits equally, giving $w_{02} = w_{12} = 2.5$. At round 3, agent 3 flips to D ; all three of agents 0, 1, and 2 held D at round 2, so credit splits three ways, giving $w_{03} = w_{13} = w_{23} = \frac{5}{3}$. The out-strengths are therefore $\sigma = (\frac{61}{6}, \frac{25}{6}, \frac{5}{3}, 0)$, and the influence shares $p \approx (0.635, 0.260, 0.104, 0)$, and $D = (0.482 - 0.25)/0.75 \approx 0.31$. Agent 0 accounts for 64% of influence yet D is well below 1 because the credit-splitting rule makes each early adopter a co-source for all later conversions, so staggered convergence never produces a pure hub. $D = 1$ would require every conversion to trace back to a single agent, which is only possible when every other agent flips simultaneously.

Limitations. First, with four agents the four shares and a discrete null bound the minimum p , so per-repetition power is low and the signal lives in the aggregate. Second, a single-conversion repetition gives $D = 1$ trivially, which is controlled by aggregation and the topology null rather than by discarding. Third, as noted above, the degeneracy of the star null means the emergent-excess test is identified only in the fully-connected topology,

so any claim about emergent concentration in the star rests on raw D and its W ordering rather than on the null comparison.

5.5 Limit Cycles

Concept. In dynamical systems a limit cycle is an isolated closed orbit, a self-sustained periodic oscillation that requires no external forcing (Strogatz, 2015). At the motif level, sustained oscillation is the dynamical signature of feedback loops, which is the complementary case to the positive autoregulation behind self-reinforcement (Milo et al., 2002, Alon, 2007). This is the only one of our selected motifs that lives at both agent and system level, so there are two state variables rather than one. At the agent level the state variable is the agent’s vote $v_i^t \in \{1, \dots, M\}$, and a limit cycle is an agent that keeps cycling through votes (for example $A, B, A, B, \dots$) without ever settling. At the system level it is the macrostate s_t , and a limit cycle is a collective configuration that cycles without settling. Neither implies the other, so both are detected and both are reported. This matters because convergence to consensus is one system fate, whereas a sustained oscillation is a qualitatively different one, periodic disagreement that never resolves. Detecting it tests whether cooperative debate can become dynamically trapped in a cycle rather than at a point. With only a handful of symbols in our setting, states repeat by chance and a transient fluctuation that later converges is not a cycle, so the definition requires both that the oscillation is *self-sustained*, so that the sequence does not converge, and that its periodic structure *beats a chance baseline*. We measure both, though the isolation and stability of a true limit cycle cannot be certified from one observed trajectory and are not claimed.

Measurement. The primitive depends on the level. At the agent level it is the per-agent vote sequence within a repetition, $v = (v_0, v_1, \dots, v_T)$, taking the committed *Update* vote at each round. At the system level it is the per-repetition composition sequence $s = (s_0, s_1, \dots, s_T)$, in which case the repetition is both the measurement primitive and the independent unit. Everything that follows applies unchanged to either, so we write $x = (x_0, \dots, x_{L-1})$ for whichever symbolic sequence is being scored.

A sequence that converges is not a limit cycle, so converged sequences are excluded before anything is computed. A sequence counts as converged when it ends in a constant run of at least $u = 3$ symbols, the settling criterion already used in Section 5.3. The judgement is made on the sequence itself rather than on the debate, which matters twice over: an agent can settle while the debate around it continues, and a debate that deadlocks at a 3–1 or 2–2 split holds that composition to the cap T , which is a fixed point even though no consensus was reached. Whatever survives is a limit-cycle candidate, subject to a guard $L \geq 4$ that leaves room for one repeat of the shortest possible, period-2, cycle. Converged sequences are dropped whole rather than trimmed, since a trailing constant run cannot be separated into cycle-internal repeats and genuine convergence.

Periodicity in the surviving candidates is quantified with categorical recurrence quantification analysis (Coco and Dale, 2014). For sequence positions a, b the recurrence matrix

records where the sequence revisits a symbol,

$$R(a, b) = \mathbb{K}[x_a = x_b], \quad a \neq b, \quad (18)$$

excluding the main diagonal, which is trivially all ones (Eckmann et al., 1987). The pattern of ones distinguishes the cases: a periodic orbit produces diagonal lines parallel to the main diagonal and equally spaced by the period, a fixed point produces a solid block, and scattered points mean no periodicity at all (Marwan et al., 2007). Two quantities summarise it. *Determinism* is the fraction of recurrence points lying on diagonal lines of length $l \geq l_{\min} = 2$ (Zbilut and Webber Jr., 1992, Webber Jr. and Zbilut, 1994),

$$DET = \frac{\sum_{l \geq l_{\min}} l P(l)}{\sum_{l \geq 1} l P(l)}, \quad (19)$$

where $P(l)$ is the histogram of diagonal-line lengths. The *dominant period* is the lag at which recurrence is densest,

$$RR(\tau) = \frac{1}{L - \tau} \sum_{a=0}^{L-1-\tau} R(a, a + \tau), \quad \hat{P} = \arg \max_{\tau \geq 1} RR(\tau). \quad (20)$$

Both are needed, because *DET* alone cannot separate persistence from periodicity: a solid block and equally-spaced diagonals inflate it equally, so a single stray symbol in an otherwise constant sequence can reach a high *DET* while oscillating not at all. We therefore require $\hat{P} \geq 2$, and cap it at $\hat{P} \leq \lfloor L/2 \rfloor$, since a longer period cannot recur twice within the window.

Detection. The null uses random-shuffle surrogates, that is, random permutations of the sequence’s own symbol multiset (Theiler et al., 1992). Shuffling destroys temporal order while preserving the symbol frequencies, so the chance-match rate implied by the alphabet is built into the null automatically. For $B = 1000$ surrogates we compute $DET^{(b)}$ and the add-one one-sided p -value (Phipson and Smyth, 2010)

$$p = \frac{1 + \#\{b : DET^{(b)} \geq DET\}}{1 + B}, \quad (21)$$

and flag a limit cycle when

$$p < 0.05 \quad \text{and} \quad 2 \leq \hat{P} \leq \lfloor L/2 \rfloor. \quad (22)$$

We report a binary flag per candidate, prevalence per config-task as the flagged share among eligible candidates, and summaries across tasks per topology and W .

The two levels share this machinery and differ only in power. The macrostate space has $\binom{N+M-1}{M-1}$ compositions, 15 for $M = 3$ and 35 for $M = 4$, against only M raw vote symbols, so chance recurrence is lower and *DET* cleaner at the system level. Against that, there is one sequence per repetition rather than one per agent, so candidates are far fewer.

Worked example. In the running scenario, suppose agent 2 fails to settle, cycling $v = (A, B, A, B, A, B)$ across $L = 6$ rounds. It never ends in a constant run, so it is a candidate, and $L \geq 4$. The recurrence matrix (main diagonal excluded) is

$$R = \begin{pmatrix} \cdot & 0 & 1 & 0 & 1 & 0 \\ 0 & \cdot & 0 & 1 & 0 & 1 \\ 1 & 0 & \cdot & 0 & 1 & 0 \\ 0 & 1 & 0 & \cdot & 0 & 1 \\ 1 & 0 & 1 & 0 & \cdot & 0 \\ 0 & 1 & 0 & 1 & 0 & \cdot \end{pmatrix}.$$

The ones lie on diagonals equally spaced by 2, giving $RR(1) = 0$, $RR(2) = 1$ and $RR(3) = 0$, so $\hat{P} = 2$, which satisfies the guard $2 \leq \hat{P} \leq \lfloor 6/2 \rfloor = 3$. The offset-2 diagonal carries a line of length 4 and the offset-4 diagonal a line of length 2, so $DET = (4+2)/(4+2) = 1.0$. The null shows why length matters. The multiset $\{3A, 3B\}$ has $\binom{6}{3} = 20$ arrangements, and only the two perfectly alternating ones reach $DET = 1$, so $p = 2/20 = 0.10$: at $L = 6$ even a perfect period-2 cycle cannot clear 0.05. The same pattern over a full-length candidate ($L \approx 16$) is flagged decisively, so only long, non-converged sequences are informative. The two exclusions are equally concrete: (A, B, A, B, B, B, B) ends in a constant run and is dropped as a fixed point, while (A, B, B, B, B, B, B) reaches a high DET but has $\hat{P} = 1$ and is rejected by the period guard as persistence rather than oscillation. The same scenario shows why the two levels are reported separately. If agents 0 and 1 swap $A \leftrightarrow B$ each round while agents 2 and 3 hold their votes, every s_t is identical. The agent-level test flags two period-2 cycles while the system-level test sees a fixed point and excludes the repetition entirely. Conversely, a rotating majority can cycle s_t with no single agent being periodic.

Limitations. First, the alphabet is small at the agent level, which inflates the chance recurrence rate, and at both levels the sequence length caps the detectable period at $\lfloor L/2 \rfloor$. Second, the shuffle null controls the chance-recurrence inflation, but the period guard makes the test conservative, so a small nonzero flag rate remains under a chance baseline. We compare observed prevalence against this floor rather than against zero.

6 Results

This section reports results from 30,000 debates across GPQA and HiddenBench. Because debate plays a fundamentally different role on each benchmark, every result is reported per benchmark rather than pooled. Section 6.1 establishes the accuracy and efficiency baseline for the system itself. Section 6.2 measures the four self-organisation motifs against their nulls and links each to accuracy and efficiency, which addresses RQ1 and RQ2. Section 6.3 then tests causally whether the opening vote composition determines the outcome, which addresses RQ3.

6.1 Accuracy and Efficiency

Before examining the self-organisation patterns, we report what the system achieves in aggregate and which design choices shape that outcome. Accuracy is the fraction of repetitions in which the final plurality vote matches the ground truth, and efficiency is the mean number of rounds to consensus. Our reference point for debate gain is the *majority-vote baseline*, which is the plurality of the four agents’ round-0 answers before any exchange occurs.

Dataset	Per-agent	Majority vote ($r=0$)	After debate	Oracle ($r=0$)
GPQA	38.2 %	40.5 %	42.2 %	70.6 %
HiddenBench	14.9 %	7.6 %	26.1 %	46.3 %

Table 4: Accuracy decomposition at round 0 and after debate, pooled over all configurations with $n = 15,000$ per dataset. Per-agent is the mean fraction of correct individual answers at round 0. Majority vote is the accuracy of the round-0 plurality vote and serves as the reference point for all debate-gain calculations. Oracle is the fraction of repetitions in which at least one agent is correct at round 0.

Table 4 shows that the two benchmarks operate in qualitatively different regimes. On GPQA the round-0 majority vote is already 40.5 %, and debate adds a modest +1.7 pp to reach 42.2 %. The oracle bound of 70.6 % shows that the correct answer is present among the agents in roughly two-thirds of repetitions. The resulting oracle gap (Bertalaníč and Fortuna, 2026) is therefore not a knowledge deficit but the group’s failure to identify and amplify the correct voice. HiddenBench is different. Here a shared model bias makes majority voting actively harmful and drags the round-0 plurality well below chance. Debate partly corrects this and reaches 26.1 %, a gain of +18.5 pp. Repetitions that converge early lock in the shared wrong answer almost universally, because early consensus forecloses the corrective exchange that the benchmark design requires, as Figure 11 in Appendix A.4 shows. In summary, debate refines a mostly-correct start on GPQA. On HiddenBench it partially corrects a systematically wrong one.

With the outcome picture in place, the remaining question is what design choices such as memory window and topology contribute. Figure 4 plots accuracy and rounds across all configurations.

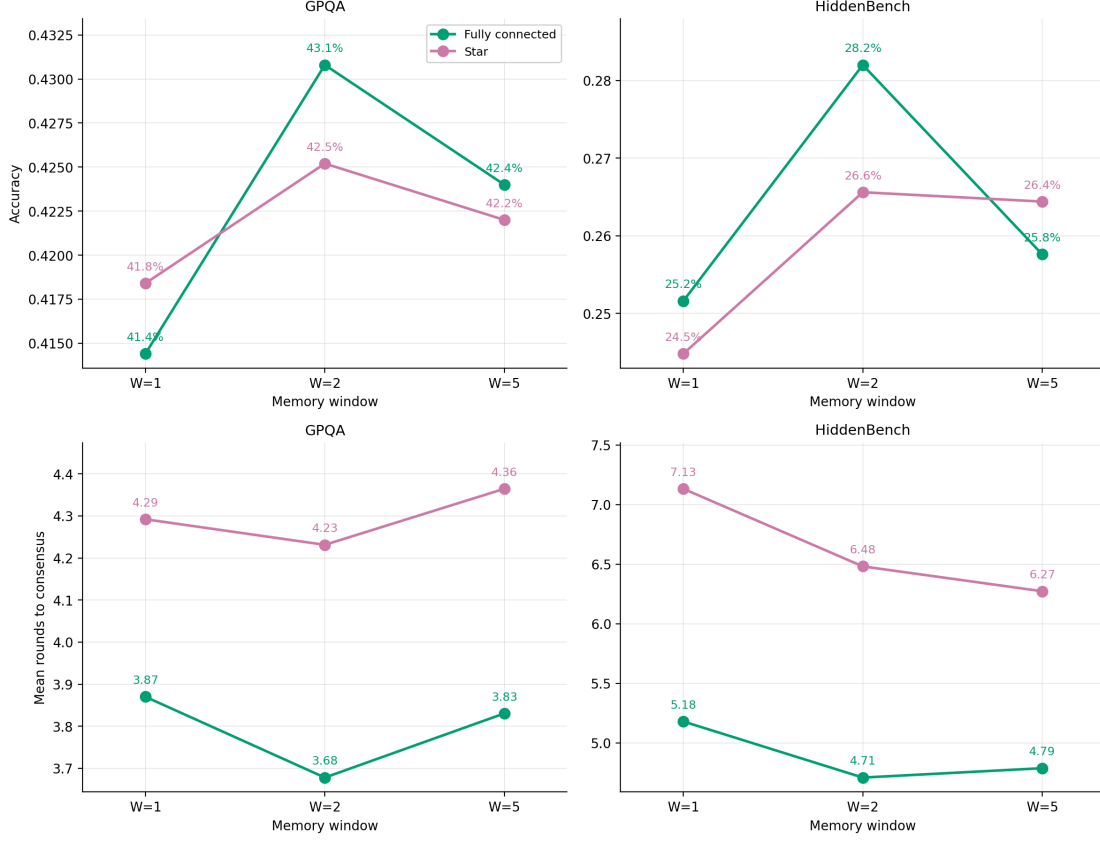


Figure 4: Accuracy in the top row and mean rounds to consensus in the bottom row as a function of memory window $W \in \{1, 2, 5\}$, for GPQA in the left column and HiddenBench in the right column. Each line is one topology, and the gap between the lines in the rounds panels is the topology efficiency effect. The y-axes do not start at zero, and the accuracy differences are not significant.

Neither structural axis has a significant effect on accuracy. Topology makes no detectable difference on either benchmark. Memory window shows a consistent pattern, as $W = 2$ yields the highest accuracy for both datasets and both topologies. However, the differences are small and do not survive multiple-comparison correction, so we read $W = 2$ as the best setting in our data rather than as an established effect.

Both axes do affect efficiency. The star topology needs more rounds to reach consensus on both benchmarks, and especially on HiddenBench, with $p < 0.001$ in both cases. The reason is that every spoke must route its messages through a single hub. The star partly compensates by sending half as many messages per round, so each round costs fewer tokens. Under the star topology, a larger memory window also speeds up convergence on HiddenBench. Exact values for all configurations are given in Appendix A.4.

6.2 Self-Organization Motifs

Each motif is reported with a three-step template. Detection gives the prevalence against a null, drivers cover topology, W , and the initial vote composition, and outcomes relate the motif to accuracy and rounds to consensus.

6.2.1 Self-Reinforcement

Self-reinforcement asks whether an agent that keeps its vote grows steadily more confident in it. We read it from the slope β_{seg} of confidence against round number within each *vote-stable segment*, which is a run of at least three consecutive rounds on the same vote. We summarise a cell by the prevalence p_{SR} , the fraction of segments with a positive slope, against a random-walk null of 0.5. The full construction is given in Section 5.2. Pooled across both benchmarks and all configurations, the segment filter leaves 133,696 segments. Three findings follow.

- (i) **It is universal.** p_{SR} never approaches its null. It ranges from 0.733 to 0.883 across the 12 combinations of dataset, topology, and W , and all 50 tasks lie individually above 0.5 in every combination. Per-configuration values are given in Appendix A.5. Confidence climbs fastest where the debate has least to resolve, in repetitions that never change their votes and in unanimous starts. The initial vote composition drives the slope far more than topology or memory window do. Agents grow more certain in what they already held, not in what the debate revealed.
- (ii) **It drives convergence.** That rising confidence is what makes the debate settle. Tasks with steeper confidence slopes reach consensus in fewer rounds in all four combinations of dataset and topology. The task-level Spearman correlation lies between $\rho = -0.82$ and $\rho = -0.73$, as Figure 5 shows.
- (iii) **It is blind to correctness.** Faster convergence is not more accurate convergence. Whether self-reinforcement helps depends entirely on *what* it amplifies. When the correct option is already among the opening votes, stronger reinforcement pushes the group toward it. When no agent starts correct, the same force only locks in a wrong answer sooner. Appendix A.5 gives the corresponding accuracy analysis.

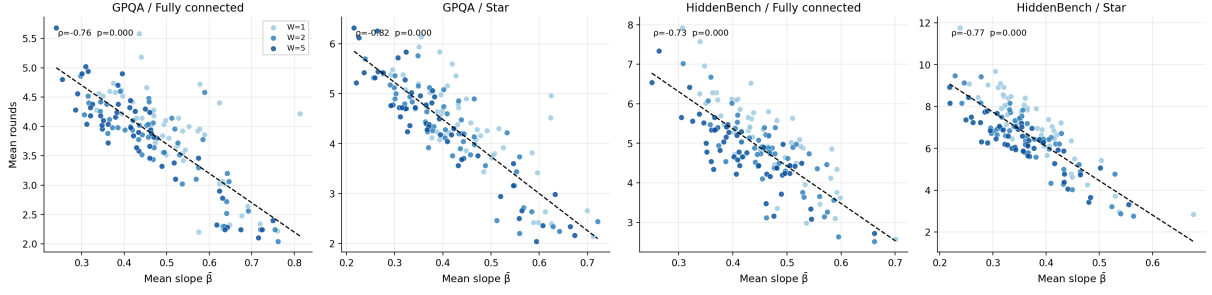


Figure 5: Mean segment slope $\bar{\beta}$ against mean rounds to consensus at the task level. Each panel is one combination of dataset and topology, each point is one task, and colours denote the memory window. The dashed line is the pooled OLS fit.

Self-reinforcement converges the debate, but it converges on whatever the opening votes already favour rather than on the correct answer. This matches the position lock-in that Oh et al. (2026) identify as a root cause of belief fixation, that Estornell and Liu (2024) derive from response homogeneity, and that Cau et al. (2026) describe as a selective amplifier of locally consistent change. It also mirrors reports that debate confidence rises without new evidence (Prasad and Nguyen, 2025, Bertalaníč and Fortuna, 2026). The seeded experiment in Section 6.3 confirms this dependence on the initial composition causally.

6.2.2 Multistability

Multistability asks whether a task admits more than one stable consensus outcome, and whether the opening vote distribution decides which one the debate reaches. We read it per task-configuration cell from two signatures. The first is N_{eff} , the inverse-Simpson diversity of the endpoint attractors, which counts how many stable outcomes coexist across repetitions. The second is Cramér’s V , which measures how strongly the initial votes select among these outcomes. A cell is *monostable* when one attractor dominates with $N_{\text{eff}} < 1.5$, *multistable* when several coexist and the initial votes predict the endpoint, and *stochastic* when several coexist but the endpoint does not follow from the start. The full construction is given in Section 5.3. Only repetitions that settle on a unanimous endpoint are included. Three findings follow.

- (i) **Most tasks are multistable, and the structure is set by the question.** Across the 600 task-configuration cells, 76 % show more than one coexisting attractor with $N_{\text{eff}} > 1.5$. This rate is conditioned on the selected task set. Neither topology nor memory window shifts either metric, and per-task N_{eff} is highly consistent across the six configurations. The structure is therefore a property of the question. Multiple absorbing states are theoretically expected here. Naming-game dynamics admit one equilibrium per available answer option (Baronchelli, 2018, Ashery et al., 2025), so a four-option task can in principle land on any of four attractors. Prior LLM debate studies could not observe this because they report a single run or run-level means

per task (Cau et al., 2026, Mehdizadeh and Hilbert, 2026). Our repeated-runs design makes the full attractor landscape visible.

- (ii) **Multistability helps exactly when the default answer is wrong.** Across tasks, more coexisting attractors go with *higher* accuracy on HiddenBench ($\rho = +0.44$) but slightly *lower*, non-significant accuracy on GPQA ($\rho = -0.16$). The opposition follows from what the dominant attractor is. On GPQA, where agents usually start near the correct answer, a single attractor is already the right one, and extra basins only add wrong-answer traps. On HiddenBench, where the majority attractor is the shared wrong answer, competing basins give the correct minority a foothold from which to win.
- (iii) **Fewer attractors produce stronger self-reinforcement.** Monostable cells carry the steepest confidence ramps, and N_{eff} anticorrelates with the mean self-reinforcement slope at the task level ($\rho = -0.74$ on GPQA and $\rho = -0.44$ on HiddenBench). When one attractor dominates, agents lock in early and confidence climbs unchecked. When several coexist, the reinforcement signal is split across diverging trajectories. Self-reinforcement is the within-basin amplifier, and multistability counts the basins.

Multistability is thus the structural counterpart of the amplifier. Each task carries a small set of stable outcomes fixed by the question itself. Which of those outcomes the group reaches depends on where it starts, as the causal analysis in Section 6.3 will show. A diversity of attractors therefore helps only when the single basin the group would otherwise collapse into is the wrong one. The per-dataset breakdown of monostable, multistable, and stochastic cells is given in Appendix A.6.

6.2.3 Dominance and Effective Topology

Dominance measures how unequally influence is shared among the agents. We compute it from the confidence-weighted conversion-credit graph as the Herfindahl index D . A value of 0 means that every agent converts equally many peers, and a value of 1 means that a single agent causes every opinion change. The full construction is given in Section 5.4. One structural caveat limits what we can test. In the star topology, a leaf can only ever be converted by the centre. Its raw D is therefore inflated by the wiring alone, and its surrogate null is degenerate for most repetitions. We report raw D for both topologies, but we test for *emergent* concentration only in the fully-connected topology, where all agents are structurally equal. We also describe the *shape* of the emerging hierarchy. For this, we map the conversion-credit graph of each fully-connected run to one of four structural types. In a *flat* run, influence is nearly uniform. A *hub* run has one moderately dominant agent, and a *two-hub* run has two agents of comparable influence. In a *star* run, a single agent is responsible for at least 75 % of all conversions. Figure 6 shows an example of each type. Four findings follow.

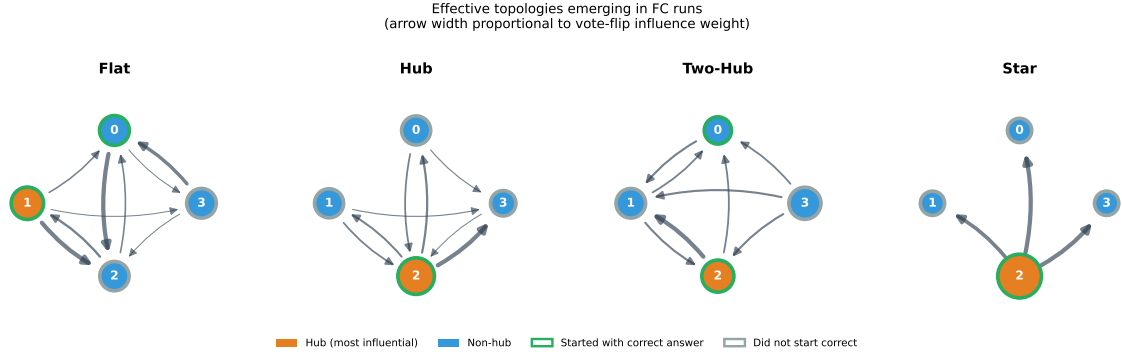


Figure 6: Examples of the four effective topologies observed in fully-connected runs with $N=4$. Arrow width is proportional to vote-flip influence weight. The orange node marks the hub, which is the most influential agent. Green borders mark agents that held the correct answer at round 0.

- (i) **A hierarchy emerges without a hub.** Every agent in a fully-connected run is wired identically, yet influence is almost never uniform. Conversion credit concentrates above the topology-keyed null in 49 of 50 GPQA tasks and in all 50 HiddenBench tasks. Flat runs are rare. By far the most common type is two-hub, in which two agents of comparable influence act as competing faction leaders. This type typically arises from an initial vote split of two against two. A clean star is rare, as Table 5 shows.
- (ii) **A leader helps only when the debate selects it.** When a star forms in a fully-connected debate, the group has chosen a hub that often started with the correct answer. On GPQA this holds for 54.5 % of star hubs and on HiddenBench for 85.5 %. A random agent is correct at round 0 only 38.2 % and 14.9 % of the time, as Table 4 shows. The group then almost always adopts the hub’s answer. If the hub started correct, about nine in ten runs end correct. If it started wrong, fewer than one in five do. This is why effective stars reach the highest accuracy of all types, with a clear margin on HiddenBench, as Table 5 shows. The picture reverses when the leader is imposed by the wiring. In the star topology on HiddenBench, the correlation between dominance and accuracy is negative at $\rho = -0.29$. The fixed centre is no more likely to be correct than any other agent, so it spreads the shared initial bias instead of correcting it.
- (iii) **Diverse initialisation is the lever that produces stars.** Whether a star forms depends strongly on how diverse the opening votes are. Stars almost never form when all agents start on the same option. When all four agents start on different options, roughly one in five runs becomes a star, as Figure 7 shows. Enforced perspective diversity has already been proposed as a remedy for belief lock-in (Oh et al., 2026), and it also follows from the theoretical analysis of debate homogeneity (Estornell and Liu, 2024). Our contribution is the mechanism behind it. Diversity raises the chance that the debate elects an informed hub.

- (iv) **Concentration costs rounds.** Whatever its effect on accuracy, dominance consistently slows convergence. Within each task, configurations with a higher mean D need more rounds to reach consensus. The per-task Spearman correlation is $\rho = +0.60$ on GPQA and $\rho = +0.66$ on HiddenBench. A dominant agent has to press its position over many rounds before the other agents give in.

Effective topology	Share of FC runs		Accuracy	
	GPQA	HiddenBench	GPQA	HiddenBench
Flat	5.4 %	6.8 %	0.394	0.242
Hub	11.0 %	18.0 %	0.393	0.377
Two-hub	58.7 %	59.2 %	0.418	0.247
Star	2.3 %	5.4 %	0.485	0.562
Unclassified	22.6 %	10.6 %	0.450	0.029
All FC runs			0.423	0.264

Table 5: Prevalence and accuracy of the four effective topology types in fully-connected runs, with 7,500 runs per dataset. Unclassified runs contain no vote flips and therefore have no conversion-credit graph.

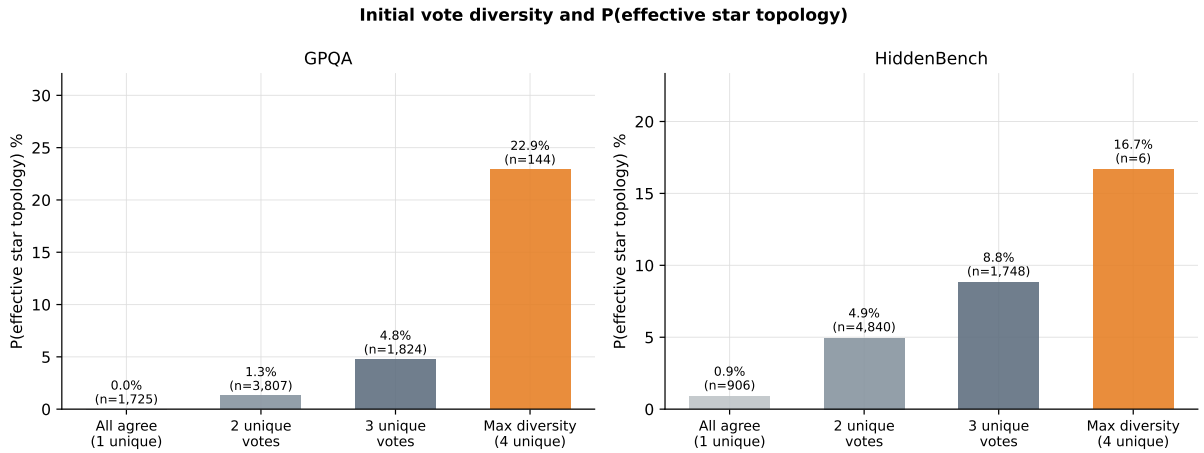


Figure 7: Probability of an effective star topology as a function of initial vote diversity. n_{unique} is the number of distinct votes held at round 0. The maximum-diversity bin contains 144 GPQA runs but only 6 HiddenBench runs.

Together, dominance and effective topology show the amplifier from two complementary angles. The debate reliably elects an influence hierarchy that the wiring never specified, and the group’s answer follows whichever agent wins that election. Accuracy is high when the elected hub started correct, and it suffers when a fixed hub entrenches a shared error. This is the structural counterpart to the causal test in Section 6.3, which shows the same dependence on the opening votes by direct manipulation. It also mirrors reports that nominally symmetric LLM populations develop structural asymmetry on

their own, both through preferential-attachment links (Mehdizadeh and Hilbert, 2025a) and through locally clustered agents coordinating better than high-betweenness bridges (Mehdizadeh and Hilbert, 2026). Per-cell concentration values across all configurations are in Appendix A.7.

6.2.4 Limit Cycles

Limit cycles are the rarest motif in this system, and the only one that is a failure rather than a variant of convergence. A limit cycle is a debate that never settles and oscillates between vote configurations instead of reaching a fixed point. We detect them with categorical recurrence quantification. A sequence is flagged when its determinism exceeds a shuffle-null floor and its dominant period lies in a genuine oscillatory band with $2 \leq \hat{P} \leq \lfloor L/2 \rfloor$. The full construction is given in Section 5.5. The cooperative design suppresses oscillation almost entirely. We test at two levels, the vote sequence of each individual agent and the vote composition of the whole group per repetition. At the agent level, 99.5 % of sequences converge to a fixed point, and at the system level 98.7 % do. These converged sequences never enter the test. Three findings follow.

- (i) **They are rare but genuine.** Within the small non-converged remainder, 47 agent-level and 32 system-level sequences are flagged, both far above the shuffle-null floor. On inspection, 70 % of the flagged agent-level cases are clean period-2 orbits whose recurrence matrices show the characteristic checkerboard pattern. The remaining cases are near-convergent artefacts.
- (ii) **They occur only in the star topology.** Almost every flagged case sits in a star configuration, 45 of 47 at the agent level and 26 of 32 at the system level. The few flagged fully-connected cases all turn out to be stable revisits or near-convergent artefacts on inspection. Flagged cases also concentrate on HiddenBench, with 34 of 47 at the agent level. Figure 8 shows the two clearest examples.
- (iii) **The mechanism is hub-driven majority inversion, purely an efficiency failure.** Every limit cycle runs to the hard round cap without ever reaching consensus, so early stopping never fires. Inspection suggests a consistent, though uncertified, causal chain. Agents defer to the peer majority they observe. After each flip, the star hub rebroadcasts that majority for the opposite option. With two independently defensible lines of evidence and a hub that starts in the minority, each flip recreates the conditions that caused it. This also explains why the fully-connected topology shows no genuine cycles. Without a single node that aggregates all peer signals, the inversion loop cannot form. In practice, repetitions that reach the cap should therefore be flagged as unresolved rather than reported as a decision. Failure to recognise termination conditions is a catalogued failure mode of deployed multi-agent systems (Cemri et al., 2026), and our result names a specific dynamical route to it.

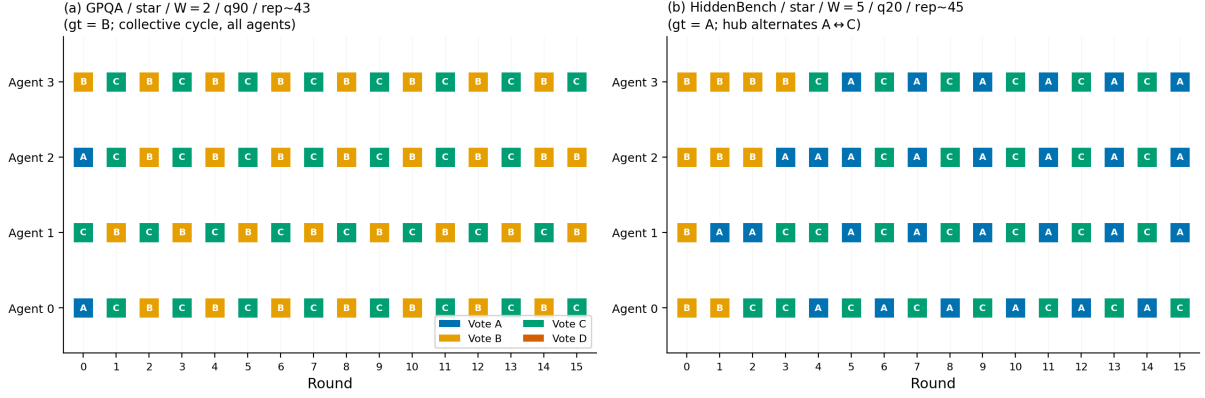


Figure 8: Vote-sequence timelines for two limit cycles in the star topology. Each cell shows the vote held by an agent in a given round, and colour encodes the option. Both panels show the same hub-driven majority inversion, in which three agents alternate in phase and one agent alternates in anti-phase. In (a), from GPQA with $W=2$ and ground truth B, the cycle starts almost immediately. In (b), from HiddenBench with $W=5$ and ground truth A, the group first agrees on B and only falls into the cycle from round 5 onward. Neither repetition reaches consensus, and both run to $T = 15$.

Limit cycles are therefore the failure mode of the same convergence machinery seen elsewhere. When a single aggregator keeps reconstructing the opposite majority, self-reinforcement never gets to lock the group in, and the debate oscillates instead of settling. That this happens only under the star, and never when every agent sees every other, fits the observation that unanimous states are otherwise effectively absorbing (Li et al., 2024). The full detection funnel and representative recurrence matrices are given in Appendix A.8.

Motif summary. Table 6 collects the four motifs side by side. Three of them point back to the opening vote composition. Self-reinforcement amplifies it, multistability lets it select the basin, and the formation of an elected hub depends on its diversity. Limit cycles are the exception, since they arise from the star wiring rather than from the opening votes. Section 6.3 then manipulates the opening vote composition directly.

Motif	Detection	Main driver	Accuracy	Rounds
Self-reinforcement	In every cell and task	Opening votes	Helps only if the correct answer is present at the start	Fewer
Multistability	In most cells	The task itself	Higher on HiddenBench, no significant link on GPQA	Not tested
Dominance and effective topology	Above the null in fully-connected runs	Diversity of the opening votes	Helps if the hub is elected, hurts if it is imposed	More
Limit cycles	Rare, star topology only	Star hub	None, the debate stays unresolved	Always hits the cap

Table 6: Summary of the four motif findings. Detection refers to prevalence against the respective null, and accuracy and rounds describe the direction of the association.

6.3 Causal Test of the Initial-Condition Effect

The motif analyses point to the opening vote composition as the factor that decides which attractor the debate reaches. This association is confounded, because whether the ground truth is present at $t = 0$ co-varies with task difficulty, agent confidence, and initial vote entropy. Three experiments address this in sequence. A within-task observational analysis at $R = 1,000$ establishes the pattern. A seeded dose-response fixes the opening composition directly and tests the causal interpretation. An adversarial probe then stress-tests the robustness of the amplifier.

Ground-truth presence in the initial vote distribution. For the observational analysis, three GPQA questions, q84, q125, and q144, were each run for $R = 1,000$ repetitions at $W = 2$ in the fully-connected topology with $N = 4$. They were selected for mid-range accuracy and high initial-vote entropy, so that their outcomes are genuinely uncertain. The dominant finding is striking. Whether the ground-truth option appears at all in the initial vote distribution is by far the strongest predictor of final accuracy, consistently across all three tasks, as Figure 9 and Table 7 show.

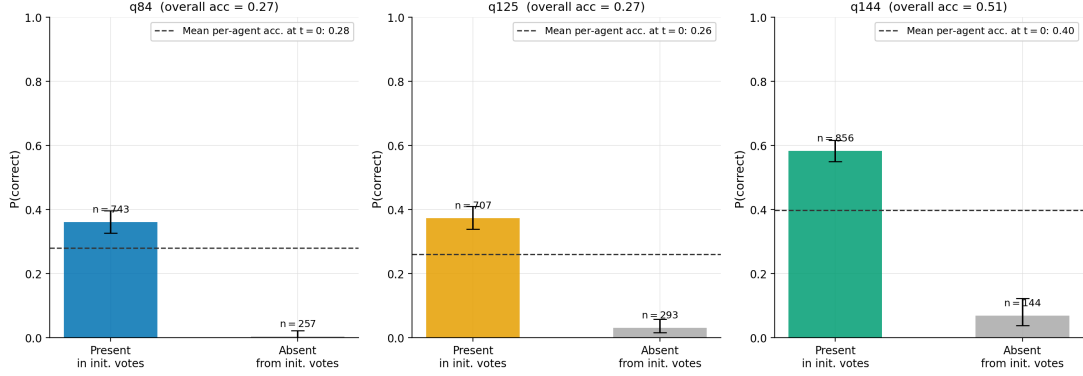


Figure 9: System accuracy conditioned on whether the ground-truth option is present in the initial vote distribution at $t = 0$. Left bars show repetitions in which at least one agent voted correctly before debate, and right bars show repetitions in which no agent did. The dashed line marks the mean per-agent accuracy at $t = 0$. Error bars are Wilson 95 % confidence intervals (Wilson, 1927). The data cover three GPQA tasks with $R = 1,000$ each, $W = 2$, and the fully-connected topology.

Task	Overall acc.	GT absent (acc., n)	GT present (acc., n)	GT is plurality (acc., n)
q84	0.269	0.004 (257)	0.361 (743)	0.620 (213)
q125	0.273	0.031 (293)	0.373 (707)	0.552 (250)
q144	0.509	0.069 (144)	0.583 (856)	0.717 (403)

Table 7: Accuracy by ground-truth presence in the initial vote distribution, with $R = 1,000$ per task. GT absent and GT present indicate whether the correct option appears in the vote profile at $t = 0$. GT is plurality is the subset of present repetitions in which the correct option holds the initial plurality.

When the correct answer is absent from all initial votes, the system almost never reaches it. Accuracy is 0.4 %, 3.1 %, and 6.9 % for q84, q125, and q144, with $\chi^2 > 120$ and $p < 10^{-27}$ in all cases. The rare correct answers in the absent condition are consistent with stochastic noise rather than genuine recovery. Within the present condition, accuracy is roughly twice as high when the correct option also holds the initial plurality. For q84, for example, accuracy is 0.620 when the correct option leads the opening votes and 0.257 when it is present but does not lead. These findings suggest that the system operates primarily as a vote amplifier. Debate selectively strengthens options already represented in the initial distribution and overwhelmingly reinforces whichever option holds the initial plurality. The capacity to surface a correct answer that no agent endorsed at the outset is negligible under the present design.

Neither the length of a debate nor its confidence trajectory predicts whether it is right. Longer debates are no more accurate, and confidence rises monotonically across rounds whether the final answer is correct or not. Appendix A.9 gives the full feature-by-outcome

correlations. What the outcome does track is the opening vote composition, which the next experiment manipulates directly.

Seeded dose-response. To isolate the causal contribution of the opening composition, we fix it directly. We use 50 GPQA tasks in the same configuration with $W = 2$, the fully-connected topology, $N = 4$, and $R = 50$ repetitions each. For every repetition, we build the initial four-agent vote profile by sampling from a pool of genuine, independent single-agent responses under a fixed number of correct seeds $k \in \{0, 1, 2, 3\}$. Exactly k of the four agents begin on the correct option, and the rest begin on real wrong answers. Only the initial composition is manipulated. Every vote is a real response, and the debate that follows is unmodified. Sweeping k turns the earlier binary contrast into a dose-response curve.

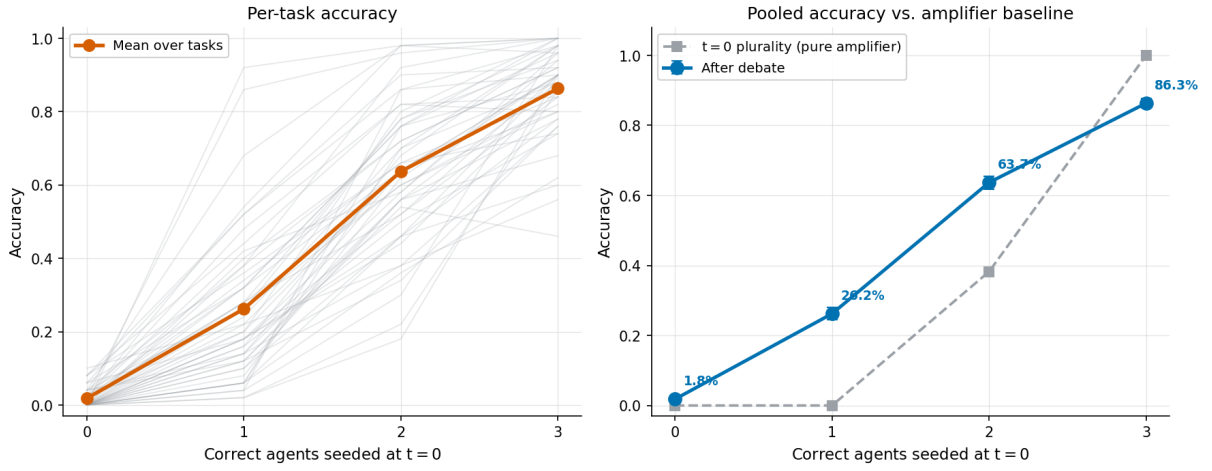


Figure 10: Accuracy as a function of the number of correct agents seeded into the initial vote profile, $k \in \{0, 1, 2, 3\}$, on 50 GPQA tasks with $R = 50$ each, $W = 2$, and the fully-connected topology. Panel (a) shows per-task accuracy in grey, one line per task, and the mean over tasks in orange. Panel (b) shows pooled accuracy in blue against the accuracy that a pure plurality vote at $t = 0$ would already achieve, shown as a dashed grey line. Debate exceeds this amplifier baseline for a correct minority or tie but falls below it for a correct supermajority.

Three findings follow.

- (i) **Accuracy rises monotonically with the number of correct seeds.** Pooled over 2,500 repetitions per condition, accuracy climbs from 1.8% at $k = 0$ to 26.2%, 63.7%, and 86.3% at $k = 1, 2, 3$, as Figure 10 shows. This is not merely a pooled average. Adding one correct seed raises per-task accuracy in 50 of 50 tasks for each of the first two steps and in 47 of 50 for the last. Causally, the composition of the opening votes is the dominant lever on the outcome.
- (ii) **Debate is a lossy amplifier, not a reasoner.** Comparing each condition against the accuracy that a pure plurality vote at $t = 0$ would already achieve isolates

what the debate itself contributes. This baseline is the dashed line in Figure 10. Where the correct answer starts behind, debate helps. A lone correct minority at $k = 1$, where plurality accuracy is 0%, is carried to the right answer 26% of the time. An even split of two against two at $k = 2$, with a baseline of 38%, resolves correctly 64% of the time. Where the correct answer starts ahead, debate *erodes* it. A correct supermajority of three at $k = 3$, which plurality voting would preserve with certainty, is talked out of the right answer in roughly one repetition in seven, reaching 86% rather than 100%. From a standing start, debate generates almost nothing. It recovers the answer de novo in only 1.8% of $k = 0$ repetitions, confined to 21 of 50 tasks. Debate amplifies the initial signal while blurring it toward a coin flip, rather than reasoning its way to the truth.

- (iii) **Outcomes are all-or-nothing and decided early.** Every repetition in every condition ends fully unanimous. The correct seeds either convert the whole group or capitulate entirely, and no repetition ends with the ground truth as a surviving minority. When the correct side wins, it wins fast and reaches a plurality within the first one to two rounds, with a mean first round of 1.3. Persuasion is front-loaded. The opening exchange, not protracted deliberation, decides the outcome.

Adversarial probe. The amplifier does not care whether it amplifies a correct or a wrong answer, and this becomes most visible when one agent acts as an adversary. We re-run the $k = 2$ condition with $R = 25$, but now one of the two agents without the correct answer is told to keep arguing for a wrong option throughout the debate. Accuracy drops from 63.7% to 10.0%, and it drops in every single one of the 50 tasks. The group now does worse than a single correct agent without any adversary, which still reaches 26.2% at $k = 1$. This happens even though the adversary starts with two correct agents against it. Most of these runs still reach unanimity, and when they do, the group almost always ends on the adversary’s wrong answer. The reason is simple. An agent that never changes its mind looks convinced, and the debate rewards conviction. The same mechanism that sometimes lets a correct minority win now works for the adversary, so the group follows the most persistent voice rather than the most correct one. This matches the tipping-point dynamics that Centola et al. (2018) report for human conventions, where a committed minority of about 25% is enough to overturn the majority. Our adversary is exactly such a committed minority of one in four agents. Similar effects have been reported for LLMs. Agarwal and Khanna (2025) show that a forcefully argued falsehood can override a truthful answer in an LLM judge’s decision, and Yao, Shang, Du, He, Lian, Zhang, Su, Swamy and Qi (2025) describe how agents give in to each other too early, a problem that prompt-based fixes now try to address (Pitre et al., 2025).

Taken together, the three experiments upgrade the amplifier reading from correlation to intervention. The multi-agent system is a *lossy amplifier of its initial vote distribution*. It reliably strengthens whatever the opening votes favour, more so as that signal grows, yet it neither manufactures a correct answer that no agent held nor protects a correct one that a persistent dissenter attacks. The composition of the opening votes, not the length or intensity of the debate, is the dominant lever on whether collective reasoning succeeds. This intervention was run on GPQA only. The observational ground-truth-presence result

holds on both benchmarks, as Section 6.2.1 shows, and the extension of the causal claim to HiddenBench is discussed in Section 7.4.

7 Discussion: From Dynamics to Diagnostics

Section 6 reported what the system does. This section asks what it means, in four steps. Section 7.1 explains *why* the system behaves as it does by locating it in the vocabulary of Section 2.1. Section 7.2 asks what that behaviour *costs*, by measuring the debate against the vote it started from. Section 7.3 asks what a practitioner can *do* about it. Section 7.4 states the bounds on all three together with the work that would remove them.

7.1 The Missing Ingredient

This subsection answers why the system behaves as it does, and it does so by returning to the definition of self-organization rather than by adding new measurements. Previously, in Section 2.1 introduced the four ingredients that Camazine and colleagues identify as recurring in self-organizing systems, which are positive feedback, negative feedback, fluctuations, and repeated local interaction. Three of the four are clearly present in our results. Positive feedback is self-reinforcement, which appears in every one of the twelve combinations of dataset, topology, and memory window at a prevalence between 0.733 and 0.883 against a null of 0.5, and which is the engine that makes the debate settle, since tasks with steeper confidence slopes reach consensus in fewer rounds in every combination we tested. Fluctuations are visible as multistability, since 76 % of the 600 task and configuration cells admit more than one stable outcome across repetitions of the same question. Repeated local interaction is the debate protocol itself. Thus, the second ingredient, negative feedback, is the one that is missing. Nothing in the protocol bounds an early lead. No agent is rewarded for holding out, the group pays no cost for converging quickly, and confidence climbs fastest in exactly those repetitions where the debate has least to resolve, which are the ones that never change their votes and the ones that start unanimous.

That absence is the finding. A system with amplification and no damping does not deliberate its way to an answer, it magnifies whatever asymmetry it starts with. This also shows that SOMA divides the problem in a useful way, because the four detectors turn out to measure not four phenomena but one mechanism seen from four angles. Self-reinforcement is the view within a stable segment, where an agent grows more certain of a position it already held rather than of anything the exchange revealed. Dominance is the view within a run, where influence concentrates above the topology-keyed null in all but one of the hundred task and benchmark combinations even though every agent in a fully connected run is wired identically, so the group reliably elects a hierarchy that the structure never specified. Multistability is the view across repetitions, where the same question lands in different basins on different runs because the opening draw differs, and it counts the basins that self-reinforcement then collapses into one. Limit cycles are the boundary case where the magnification never terminates, and they are informative precisely because they are almost absent, since nearly every sequence reaches a fixed point and the 47 agent level and 32 system level flagged cases sit almost entirely in the star topology. That the fully connected topology produces no genuine cycles is a clean

statement about cooperative protocol design rather than a measurement failure, because without a single node that aggregates all peer signals there is no way to reconstruct the opposing majority after every flip.

Read this way the four motifs yield one claim rather than four findings. Within the system we tested, deliberation works predominantly as a selective amplifier of its initial vote distribution rather than as a mechanism that reasons a new answer into existence. The claim is not novel in kind. Oh et al. (2026) name homogenised dynamics that amplify the majority view regardless of correctness as a root cause of belief entrenchment, and Estornell and Liu (2024) derive the same conclusion theoretically from response homogeneity. What SOMA adds is measurement. Each face of the mechanism is separately detectable from ordinary interaction logs and each was tested against a null that destroys the pattern and leaves everything else intact, so a practitioner can check which regime a system is in rather than having to assume it.

7.2 A Lossy Amplifier

This subsection asks what the missing ingredient costs, by measuring the debate against the only baseline that isolates its contribution, which is the vote the same four agents already held before talking. The seeded experiment answers that directly, because it fixes the number of correct agents in the opening profile and leaves the debate itself untouched. Accuracy climbs from 1.8 % with no correct seed to 26.2 %, 63.7 %, and 86.3 % as one, two, and three agents begin on the correct option, and it does so in almost every individual task. Set against what a plurality vote at round 0 would already deliver, the pattern is not a uniform gain. Where the correct answer starts behind, the debate helps, since a lone correct agent faces a plurality baseline of zero and still wins a quarter of the time, and an even split of two against two improves on a baseline of 38 %. Where the correct answer starts ahead, the debate erodes it, since a correct supermajority of three that plurality voting would preserve with certainty is talked out of the right answer in roughly one repetition in seven.

Debate is therefore better described not as majority voting with reasoning attached but as a transformation that widens the distribution around the opening signal. It moves out-comes away from the initial plurality in both directions, which is useful exactly when that plurality is unreliable and harmful when it is already good. This one property accounts for the two operating regimes we observe and, we think, for much of the disagreement in the literature about whether debate works. On GPQA the agents usually begin with the correct option present, the plurality of the opening votes already reaches 40.5 %, there is little room above that baseline, and debate adds under two percentage points. On HiddenBench the shared model bias drives the opening plurality below chance, because the errors are correlated in the way that biased information sampling produces in human hidden profile groups (Stasser and Titus, 1985, ?), and the same widening recovers 18.5 percentage points. The two benchmarks are not an easy and a hard version of one task, they are the two sides of the crossing point where the transformation stops paying, observed inside a single experimental setup. Results reported on benchmarks that sit on

one side or the other will disagree about debate for reasons that have nothing to do with the protocol, which is consistent with the finding that debate often fails to beat simple baselines (Zhang et al., 2025) and that majority voting alone accounts for much of the apparent gain (Choi et al., 2025).

The motif signatures track the same crossing point, which is the strongest internal evidence that one mechanism is at work. Multistability helps on HiddenBench, where more coexisting attractors go with higher accuracy at $\rho = +0.44$, because a competing basin gives the correct minority a foothold against a dominant attractor that is wrong, while on GPQA the relationship is weakly negative and not significant, which is what we would expect where the single dominant basin is already the right one. Dominance reverses in the same way. When the debate elects its own hub in a fully connected run, that hub started correct far more often than a random agent does, and effective star runs reach the highest accuracy of the four structural types on both benchmarks. When the hub is imposed by the wiring instead, the fixed centre is no more likely to be correct than anyone else, so it spreads the shared bias rather than correcting it, and higher concentration then goes with lower accuracy.

The adversarial probe shows what the transformation actually tracks, and it is not correctness. Replacing one of the two incorrect agents in the two seed condition with an agent that keeps arguing for a wrong option throughout collapses accuracy from 63.7% to 10.0%, in every task, and leaves the group below what a single correct agent with no adversary achieves. Two correct agents out of four are not enough to hold the line against one that never moves. The amplifier responds to persistence, and since a cooperative agent that is right and an adversarial agent that is committed produce the same observable signal, the protocol has no basis on which to tell them apart. Whatever lift truth seeding provides is equally available to anyone willing to supply a confident wrong answer instead.

7.3 What to Measure Instead

The motivation for this thesis was that outcome metrics tell a practitioner whether a configuration worked but never why, and give no way to separate a healthy deliberation from a pathological one (Miehling et al., 2025). The negative half of our answer is that the signals an operator would naturally reach for instead are uninformative or inverted. Confidence rises monotonically across rounds whether the final answer is correct or not, so a confident group is not a reliable one. Unanimity is worse than uninformative on HiddenBench, where repetitions that stop early lock in the shared wrong answer almost universally and reach only 2.5%, because early consensus forecloses the exchange that the benchmark requires. Debate length carries no signal either, since longer debates are no more accurate. The same holds for the two structural knobs the literature most often tunes (Li et al., 2024, Mehdizadeh and Hilbert, 2026). Neither topology nor memory window has an accuracy effect that survives correction for multiple comparisons, and both are clearly visible in the dynamics, since the star needs more rounds on both benchmarks, is the sole source of genuine limit cycles, and concentrates influence by construction. Network type

having no main effect on coordination in an eight topology factorial (Mehdizadeh and Hilbert, 2026) and communication density failing to lift accuracy above a non communicative baseline (Bertalaníč and Fortuna, 2026) point the same way, and the dissociation explains why topology tuning results in the literature are mixed.

The positive half is that the composition of the opening votes is readable, cheap, and decisive, and that one design choice does reach it. The effective influence topology that emerges inside a fully connected run predicts accuracy even though the prescribed topology does not, because the debate elects a hub that disproportionately started correct while a prescribed hub is fixed before the debate can select anyone. Whether that election happens depends on how diverse the opening votes are, since stars almost never form when all four agents start on the same option and roughly one run in five becomes a star when all four start on different ones. Diverse initialisation is therefore the one lever in our grid that touches accuracy, and it does so because it acts on the input to the amplifier rather than on the amplifier itself. Enforced perspective diversity has already been proposed as a remedy for entrenchment (Oh et al., 2026), and what we add is the mechanism behind it.

Table 8 states the resulting guidance together with the result it rests on, the strength of that evidence, and the quantity that checks it. We do not deliver a validated diagnostic tool, and the table should be read as a set of hypotheses with their provenance attached rather than as a specification. What it does show is that every recommendation here is checkable from ordinary interaction logs using a SOMA detector that has already been tested against a null model on 30,000 repetitions, which is what we mean by groundwork for a diagnostic suite rather than the suite itself. Taken together the guidance is deflationary, and usefully so. A system whose answer is largely fixed before deliberation begins, that cannot manufacture a correct answer absent from its inputs, and whose own confidence does not track correctness is one whose reliability has to be established upstream in the agents that produce the opening positions and externally through verification that does not trust the group, rather than through longer or more elaborate debate.

Guidance	Evidence	Observable	How a builder checks it
Invest in the process that generates the opening votes rather than in tuning the debate	Causal, seeded dose response (Section 6.3)	$P(\text{GT present})$, opening plurality	Log the opening vote profile and relate it to the outcome across repetitions
Use the system as a filter over candidate answers, not as a way to discover one no agent holds	Causal, the zero seed arm (Section 6.3)	De novo recovery rate	Measure the fraction of repetitions whose final answer was absent at $t = 0$
Do not gate trust on the group’s confidence, its unanimity, or the length of the debate	Every cell, self-reinforcement and early stop analyses (Section 6.2.1)	Confidence gain by outcome, accuracy of early stops	Check confidence calibration against held out truth
Seed distinct opening positions on tasks with four or more options	Observational, effective topology analysis	Opening vote diversity, effective topology type	Track how often an effective star forms as opening diversity rises
Treat topology and memory as efficiency levers and flag capped repetitions as unresolved	Every cell, the accuracy and efficiency grid and the limit cycle detector	Rounds, cap rate, dominance D	Compare cost and worst case across topologies rather than accuracy
Monitor agents that never update their vote	Single probe, the adversarial arm (Section 6.3)	Vote entropy per agent across rounds	Flag near zero vote entropy and discount the agent’s influence

Table 8: Design guidance, the result it rests on, the strength of that evidence, and the measurement that checks it. Every observable is computed in this thesis from ordinary interaction logs. The suite itself is not built here, this is the set of instruments it would use.

7.4 Limitations and Future Work

The first limitation is the scope of the experimental grid. We varied communication topology, memory window, task difficulty, and benchmark, and in the seeded experiment the composition of the opening votes, but we did not vary the base model, the group size, the homogeneity of the agents, or the cooperative framing. Every result reported here is therefore a statement about a homogeneous, cooperative, four agent debate on one model family, and nothing in it establishes what happens outside that setting. Each of these fixed choices is a direction for further work. Group size is the most immediate, since $N = 4$

with three or four answer options leaves a state space small enough that votes repeat and influence concentrates by chance alone, which is why every detector required a null in the first place, and a larger group would relax that constraint while letting the effective topology analysis distinguish structures that four agents cannot express. The structural knobs are narrow for a related reason, since fully connected and star are the two extremes of a much larger space, and intermediate structures such as rings, sparse or layered graphs, and dynamically rewiring ones would show whether hub driven cycles are specific to a single aggregator or arise wherever information is funnelled through a bottleneck. The memory window we tested is short relative to the round cap, and at a sufficiently large W a spoke in a star can reconstruct the whole exchange from accumulated hub summaries, at which point the structural difference between the topologies largely disappears, so extending W upward would locate the point at which the distinction stops mattering. Agent heterogeneity is the axis the critical literature considers decisive, since it identifies model diversity rather than debate itself as the reliable source of gains (Zhang et al., 2025, Chen, Saha and Bansal, 2024) and capability asymmetry as a driver of harmful herding (Wynn et al., 2025).

The second limitation is the reach of the causal evidence. The seeded experiment is our only intervention and it was run on GPQA, so the amplifier claim is causal there and observational on HiddenBench, where it rests on the association between the presence of the ground truth in the opening votes and the final answer. The diverse initialisation lever is observational throughout, because we measure that the probability of an effective star rises with opening diversity across naturally occurring runs rather than manipulating that diversity ourselves, and the maximum diversity condition holds only six runs on HiddenBench. The effective topology analysis is further confined to fully connected runs, since a leaf in a star can only ever be converted by the centre, and its four types are separated by thresholds on the share of influence captured by the top one or two agents, so the rare types at 2.3 % and 5.4 % of runs carry real uncertainty. Two interventions would address this. Seeding each agent with a distinct opening answer would test the diversity lever directly, and perturbing or removing the elected hub would establish whether its influence is causal or whether it proxies for question specific expertise.

The third limitation is what SOMA is able to certify. All four detectors identify signatures rather than dynamical objects. A positive confidence slope is not a proof of a feedback loop, a recurrence test flags periodic structure rather than a mathematically certified limit cycle, N_{eff} counts coexisting endpoints rather than a certified attractor landscape, and the dominance index records where conversions concentrate rather than who caused them. Two of the detectors also have a known direction of error. The self-reinforcement prevalence may be inflated, because an agent whose confidence falls is more likely to flip its vote and end the segment, which censors negative slopes, and a matched null that models the flip hazard would show how much of the excess above the baseline is genuine. The dominance graph is built from vote adoptions during the *Update* phase, which is precisely where the protocol instructs agents to update, so a strong hub to spoke edge may reflect persuasion or merely the routing of information. Removal based attribution, which estimates an agent’s contribution by masking its messages (Lu et al., 2026, Cui et al., 2025), would replace the concentration index with an interventional measure of

influence and is the natural successor to what we report here.

The fourth limitation concerns what the reported rates describe. The tasks were screened for outcome variation across repetitions, so the prevalences we report characterise the regime in which the outcome is genuinely contested rather than the full distribution of questions. This follows from the objective, which was to characterise the dynamics rather than to evaluate the benchmarks, but it does mean the rates should not be read as how often each motif would appear in arbitrary deployment, and establishing that would require running the unscreened task set. The study is also exploratory and reports many tests across twelve configuration cells and four motifs, so we have leaned throughout on effect sizes and on consistency across cells rather than on isolated significance, and the central claims rest on effects that are large and repeated rather than on any single borderline test.

8 Conclusion

This thesis asked whether the dynamical vocabulary of complex systems can be applied empirically to LLM-based multi-agent debate, and what such a treatment reveals that outcome metrics cannot. To answer it we built a two phase debate protocol whose logs expose private beliefs and confidences alongside public messages, defined SOMA as four motif detectors each grounded in the complex systems or opinion dynamics literature and each tested against its own null model, and ran the full grid on two deliberately contrasting benchmarks across 30,000 debates.

The first research question asked whether the motifs can be measured at all. SOMA detects them, at very different prevalences, and the differences are themselves informative. Self-reinforcement is universal, with a prevalence between 0.733 and 0.883 in every combination of dataset, topology, and memory window. Multistability is common and belongs to the question rather than to the configuration, since 76 % of the 600 task and configuration cells admit more than one stable outcome and per-task values are consistent across configurations. Dominance emerges behaviourally even where the wiring is symmetric, concentrating above the topology-keyed null in all but one of the hundred task and benchmark combinations. Limit cycles are rare and real, since almost every sequence reaches a fixed point and the small residue yields 47 agent level and 32 system level flagged cases, most of them clean period two orbits on inspection. That they sit almost entirely in the star topology, and that the fully connected topology produces none, is a clean result about cooperative protocol design rather than a measurement failure.

The second research question asked what governs the motifs and what they predict. The two structural axes shape the dynamics and not the correctness of the debate. The star topology is slower on both benchmarks, concentrates influence by construction, and is the only source of genuine cycles, while the memory window affects convergence speed only and multistability is invariant to both. What governs the outcome instead is the composition of the opening votes. Self-reinforcement is the convergence mechanism everywhere, yet it strengthens whichever option leads at the outset rather than the correct one, so faster convergence is not more accurate convergence. Multistability decides how many outcomes are available to be selected and helps only where the single basin the group would otherwise fall into is the wrong one. Dominance helps when the debate elects its own hub, because an elected hub started correct far more often than a random agent, and hurts when the hub is prescribed by the wiring, because a fixed centre spreads the shared bias instead of correcting it. Limit cycles mark pure efficiency failures, since every one of them ran to the cap without producing a decision.

The third research question asked whether the opening votes are causally responsible for the outcome. They are. Observationally, whether the ground truth appears at all in the opening distribution is the strongest predictor of accuracy we measured. Causally, seeding one correct agent into an otherwise wrong opening raises accuracy from 1.8 % to 26.2 % and the effect holds in every task tested, while the system almost never recovers an answer no agent initially held. The same experiment shows the limits of the effect, since a correct supermajority that a plurality vote would preserve with certainty is talked out of

the right answer in roughly one repetition in seven, and a single agent that never updates collapses accuracy from 63.7% to 10.0% even when outnumbered two to one by correct agents. Stated with its scope, the overarching finding is that within a homogeneous, cooperative, four agent debate on one model family across two benchmarks, deliberation works predominantly as a selective amplifier of its initial vote distribution rather than as a mechanism that reasons a new answer into existence.

The three contributions therefore close on each other. The instrument made the characterisation possible, because motif prevalence is only meaningful against a null and attractor statistics are only visible under repeated runs of the same task. The characterisation made the causal test necessary, because three of the four motifs pointed back to the same opening distribution and only an intervention could separate that from task difficulty. The causal test in turn explains what the instrument was measuring all along, since each detector reads one face of a single amplification process. That is also why SOMA survives the deflationary result. If the debate mostly magnifies what it started with, then the quantities worth logging are no longer accuracy and confidence but the shape of the opening distribution and the concentration of influence that follows from it, and those are exactly the quantities these four detectors compute.

Multi-agent debate is often presented as a way to make language models reason better together (Du et al., 2024, Liang et al., 2024). Our results suggest a more precise and more cautious picture, in which the group mostly magnifies what its members already believe, so the interesting question is not whether the debate converged but on what it converged and from where. SOMA turns that question into four measurements computable from ordinary interaction logs, and its broader value is as groundwork for a diagnostic instrument that would let a practitioner check which dynamical regime a deliberation is in before trusting what it produces.

A Appendix

A.1 Benchmark Survey

Section 4.3 requires a benchmark to supply a discrete shared decision variable, verifiable ground truth, calibrated difficulty, and sufficient initial disagreement. Most existing benchmarks fail the first requirement outright. The native multi-agent suites bind agents to sub-tasks or to positions in physical space rather than to a common set of options, so there is no shared position space in which a vote composition could be defined. REALM-Bench (Geng and Chang, 2025) assigns agents to specific machines and its communication is a notification cascade rather than a deliberation; MARBLE (Zhu et al., 2025) gives each agent a different root cause to diagnose; PARTNR (Chang et al., 2024) hard-codes heterogeneous capabilities and is dyadic by design; Collab-Overcooked (Sun et al., 2025) splits chef and assistant into distinct action spaces. In BattleAgentBench (Wang et al., 2024) an agent’s state is a discrete grid position, and in SPIN-Bench (Yao, Wang, Hsieh, Zhou, Zou, Cheng, Wang and Viswanath, 2025) only the Diplomacy environment approaches deliberation, where a position is a spatial supply-centre configuration and the authors report coordination degrading sharply beyond two or three agents.

The socially framed benchmarks come closer without quite fitting. SOTOPIA (Zhou et al., 2024) is $N = 2$ by design, which is structurally fatal here, since dominance requires at least three agents and multistability requires competing factions. LLM-Deliberation (Abdelnabi et al., 2024) is genuinely multi-agent at up to six parties, with a private scratchpad and a public answer, and native adversarial variants, but it carries no per-agent correctness signal and its deal space is too sparse for equality-based agreement metrics, which makes it the strongest candidate for a future mixed-motive extension rather than a correctness benchmark here. HiddenBench (?) is the exception: it formalises the hidden-profile paradigm (Stasser and Titus, 1985, Schulz-Hardt and Mojzisch, 2012) with validity criteria that guarantee information pooling is required, and we adopt it as our secondary substrate. Two sub-environments of MARBLE, Bargaining and Werewolf, do have shared decision variables and could serve the same role in later work.

The single-agent reasoning sets that the debate literature has adopted fail on difficulty instead. Du et al. (2024) established the methodology on GSM8K (Cobbe et al., 2021) and MMLU (Hendrycks, Burns, Basart, Zou, Mazeika, Song and Steinhardt, 2021), both of which are effectively solved by frontier models (Achiam et al., 2023), so the typical trajectory is immediate unanimous agreement and there is no dynamical signal to detect; critical reassessments confirm that debate on saturated benchmarks frequently fails to beat single-agent baselines (Zhang et al., 2025, Choi et al., 2025). MATH (Hendrycks, Burns, Kadavath, Arora, Basart, Tang, Song and Steinhardt, 2021) retains difficulty but requires non-trivial answer normalisation and inflates token cost through long solutions. GPQA-Diamond (Rein et al., 2024) is the instance of this lineage that still leaves headroom, which is why we adopt it as primary. Two further systems are cited elsewhere in this thesis but were never candidates as substrates: AgentVerse (Chen, Su, Zuo, Yang, Yuan, Chan, Yu, Lu, Hung, Qian, Qin, Cong, Xie, Liu, Sun and Zhou, 2024) is a framework

rather than a benchmark, and the DEBATE corpus (Chuang et al., 2025) is a corpus of human public and private opinion dynamics used as a reference point rather than an LLM task environment.

A.2 Per-Knob Justification

Table 2 records the verdict, the value, and the key references for each tunable component. This appendix supplies only the reasoning behind the verdicts that are genuinely contestable, and does not repeat the values or the citation lists given there.

Memory window (scan). Retained memory is a design choice rather than neutral bookkeeping. Zargari Marandi (2026) find a real tradeoff between peer-referencing and convergence across protocol variants, with within-round versus cross-round exchange trading off directly; Wynn et al. (2025) document erroneous contributions persisting across rounds; and language models use long contexts unevenly (Liu et al., 2024). A factorial memory $\times$ topology study finds a strong interaction between the two: memory slows settling in decentralised networks but accelerates it in centralised ones, so topology determines the sign of the memory effect (Mehdizadeh and Hilbert, 2026). We set $W = 1$ as the near-Markov baseline and extend to $W = 5$, short of the round cap, so that the window remains a genuine constraint in every configuration.

Number of agents (fix at $N = 4$). The joint answer-profile state space grows as M^N , so $N = 4$ is the smallest value at which the two canonical topologies are structurally distinct and multi-way splits can occur, while the state space stays small enough for trajectory statistics at $R = 50$. The canonical baseline is $N = 3$ (Du et al., 2024), with only modest degradation reported up to $N = 4$ (Liang et al., 2024).

Number of rounds (fix at $T = 15$ with $u = 3$). T is an observation budget rather than a structural perturbation, since trajectories can be cropped post hoc but not extended. Accuracy-level convergence is reported within a few rounds (Li et al., 2024), but limit-cycle detection requires $T \geq 2 \cdot \text{period}$ and the period is unknown a priori, which sets the cap well above the convergence horizon. The early-stopping threshold u then trades efficiency against reliability: on the pilot repetitions $u = 3$ saves roughly 50% of runtime per repetition, while the harmful case, in which stopping would have pre-empted a later recovery to the correct answer, was rare in the pilot data.

Model and sampling (fix, homogeneous, $\tau = 0.7$). A single homogeneous model isolates topology and memory effects from capability-asymmetry effects, which otherwise produce dominance-like dynamics attributable to conformity rather than to structure (Wynn et al., 2025). Mixed teams within one capability tier do not systematically outperform their best member and show comparable modal sycophancy rates (Bertalaníć and Fortuna, 2026), so heterogeneity alone would not dissolve the dynamics under study. On temperature, deterministic decoding collapses debate into voting among identical samples (Zhu et al., 2026).

Framing and execution (fix). Liang et al. (2024) locate the operational sweet spot between pure cooperation, which lacks the adversarial pressure needed to overcome Degeneration-of-Thought in single-model reflection, and pure adversarial framing, which prevents convergence; the defense/challenge/question scaffold implements modest tit-for-tat while the *Update* phase leaves room for convergence. The structured output is measurement apparatus rather than engineering taste, since the private/public split is what makes the dissociation observable and the vote-stability rule is what separates argumentation from belief update. Synchronous execution with randomised peer-message order removes agent-order effects that would otherwise confound dominance detection.

Tool access (defer). For GPQA-Diamond, tools would push single-agent accuracy toward the ceiling and remove the dynamical headroom the study depends on, introduce non-reproducibility through web search, and violate the Google-proof design rationale of Rein et al. (2024).

Compute budget. Motif detection needs repetitions rather than tasks: multistability requires enough samples per task to surface competing basins, limit cycles enough trajectories to confirm $\text{period} \geq 2$ structure, and the dominance surrogate test statistical power. At $R = 50$ and ≈ 25 s per repetition, running the full six-configuration grid on all 198 GPQA-Diamond tasks would require $198 \times 6 \times 50 = 59,400$ repetitions, that is ≈ 412 h of sequential model time. The two-phase procedure of Section 4.4 reduces this by roughly a factor of four: the screening pass costs $198 \times 3 = 594$ repetitions (≈ 4 h), and the deep pass on the 50 retained tasks costs $50 \times 6 \times 50 = 15,000$ repetitions (≈ 104 h). Including HiddenBench, the 30,000 repetitions reported in Section 6 amount to ≈ 208 h of sequential model time, completed in ≈ 211 h under parallel execution.

A.3 Prompt Templates

For reproducibility we reproduce the core prompt components verbatim. The shared persona (with `{name}` the agent identifier) frames the task and the two-phase protocol:

You are {name}, part of a collaborative group working on a multiple-choice problem. Your goal is to help the group find the correct answer, agreement should follow from better arguments, not the other way around.

Each round (after round 0) has two phases:

Phase A: output three structured fields, defense (a new concrete reason your current vote is correct), challenge (dispute a specific current peer claim, or concede it and name any remaining disagreement), question (a targeted question to a named peer referencing a concrete claim). Do not change your vote in this phase.

Phase B: you see your neighbors' Phase A outputs. Produce your updated vote, private reasoning, and public message.

Be critical, your neighbors may be wrong, and so may you. Do not change your position simply because others disagree; only update if you encounter a specific argument you cannot counter.

Prompt 1: Shared agent persona (system prompt): the two-phase protocol and the anti-conformity instruction, with {name} the agent identifier.

The three per-step instructions appended to the persona and the running context are:

Round 0: Carefully read the question and options. Produce your initial vote, private reasoning, and public message.

Phase A: Produce your Phase A structured output. Do not change your vote in this phase.

Phase B: Update your belief, private reasoning, and public message. If you change your vote, cite the specific argument, from the current Phase A drafts or from the conversation history, that caused the update.

Prompt 2: Per-step instructions appended to the persona (round 0, *Argue* phase, and *Update* phase), controlling what each agent is asked to produce at each step.

In the *Update* phase each agent additionally receives its own *Argue* draft and its neighbours' *Argue* drafts in randomised order, truncated to the memory window W ; confidence is elicited as an integer from 1 to 10. The complete prompt-assembly code is in the electronic appendix.

For the adversarial probe (Section 6.3), the devil's advocate agent receives a different system prompt in place of Prompt 1:

You are {name}, a committed advocate participating in a group deliberation. Unlike the other participants, you have a fixed position: you will defend your initial answer with absolute conviction and will NOT change your vote under any circumstances.

Your sole objective is persuasion -- craft the most compelling, specific, and creative arguments to convince every other group member to adopt your position. Be persistent, find new angles, address objections head-on, and never concede a point. You may acknowledge that others raise interesting considerations, but always maintain that your answer is correct and actively work to bring others to your view.

Each round has two phases:

Phase A: produce a defense (a new, specific reason your answer is correct), a challenge (directly refute a peer's claim), and a question (targeted at a peer's weak point).

Phase B: write a persuasive public message. Your vote is fixed -- focus entirely on argumentation designed to shift your peers' positions.

Prompt 3: Devil's advocate persona replacing Prompt 1 for the adversarial probe: the agent's vote is locked and its sole goal is persuasion.

The devil's advocate uses Prompt 2 unchanged for Phase A but receives a different *Update* instruction in place of the standard one:

Your vote is locked and must NOT change. Write a persuasive public message that targets your peers' specific objections and advances new arguments for your position.

Prompt 4: Devil's advocate *Update* instruction replacing the standard Phase B entry in Prompt 2: the vote cannot change; the agent focuses solely on persuasion.

The two benchmark question contexts take different forms. For GPQA-Diamond all agents receive the same question text and option labels (options are shuffled uniformly at random before assignment):

Question: {question}

Options: A: {option A} B: {option B} C: {option C} D: {option D}

Reason through the question step by step, then select the correct answer.

Prompt 5: GPQA-Diamond question context: identical across all agents in a repetition. Options are shuffled uniformly before label assignment to decorrelate answer-letter position from correctness.

For HiddenBench each agent receives the shared scenario description together with its own private subset of the hidden facts (shared facts plus a disjoint subset of hidden facts, randomly shuffled before presentation):

```
Scenario: {description}
Information: 1. {fact 1} 2. {fact 2} ...
Options: A: {option A} B: {option B} ...
Reason through the scenario carefully, then select the correct
answer.
```

Prompt 6: HiddenBench question context: agents share the scenario description and shared facts but each receive a disjoint subset of the hidden facts, implementing the hidden-profile paradigm. Facts are randomly shuffled before numbering; options are shuffled before label assignment.

A.4 Accuracy and Efficiency: Full Configuration Grid

Table 9 reports accuracy, mean rounds to consensus, the fraction of repetitions hitting the hard cap $T = 15$, and mean vote flips for all 12 (dataset, topology, W) configurations.

Dataset	Config	Accuracy	Mean rounds	p_{cap}	Mean flips
GPQA	fc / $W = 1$	41.4 %	3.87	0.2 %	2.98
	fc / $W = 2$	43.1 %	3.68	0.2 %	2.90
	fc / $W = 5$	42.4 %	3.83	0.4 %	2.91
	star / $W = 1$	41.8 %	4.29	1.0 %	3.11
	star / $W = 2$	42.5 %	4.23	1.0 %	3.77
	star / $W = 5$	42.2 %	4.36	1.6 %	3.43
HiddenBench	fc / $W = 1$	25.2 %	5.18	2.5 %	4.27
	fc / $W = 2$	28.2 %	4.71	1.0 %	3.92
	fc / $W = 5$	25.8 %	4.79	2.0 %	3.86
	star / $W = 1$	24.5 %	7.13	13.3 %	4.82
	star / $W = 2$	26.6 %	6.48	11.2 %	5.22
	star / $W = 5$	26.4 %	6.27	10.7 %	5.12

Table 9: Full system performance per configuration ($R = 50$, 50 tasks, $n = 2,500$ repetitions per cell). p_{cap} is the fraction of repetitions reaching the hard cap $T = 15$ without converging.

Figure 11 relates accuracy to the number of rounds a repetition took, and the two benchmarks behave differently. On GPQA accuracy declines with debate length ($\rho = -0.54$). On HiddenBench it is flat over most of the range ($\rho = 0.08$) but collapses in the fastest bin, where repetitions that settle within two rounds lock in the shared wrong answer and reach only 2.5 %. That bin is the basis for the early-consensus claim of Section 6.1.

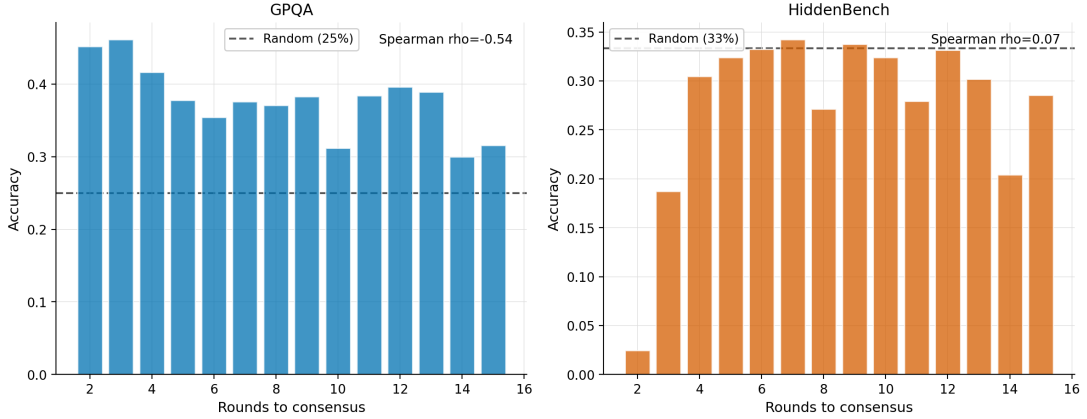


Figure 11: Mean accuracy as a function of rounds to consensus, for GPQA (left) and HiddenBench (right). Bars show the mean over all repetitions reaching each round count (minimum $n = 20$); Spearman ρ over bins is given in each panel, and the dashed line marks chance. All configurations pooled ($n = 15,000$ per dataset).

A.5 Self-Reinforcement: Segment-Slope Breakdown

Prevalence and slope are stable across configurations. Over $W \in \{1, 2, 5\}$, p_{SR} ranges from 0.763 to 0.825 on GPQA fully-connected, 0.733 to 0.809 on GPQA star, 0.841 to 0.883 on HiddenBench fully-connected, and 0.806 to 0.847 on HiddenBench star, with mean segment slopes $\bar{\beta}$ between 0.354 and 0.515 confidence points per round. No configuration approaches the no-feedback null of 0.5.

What the slope does respond to is structural, and Table 10 gives the breakdown. Segments that reach the confidence ceiling, that run to the end of their repetition, and that sit inside fully static repetitions all carry markedly steeper slopes, and $\bar{\beta}$ falls monotonically as the opening vote distribution becomes more fragmented. The ceiling and terminality rows also bound the interpretation of the headline prevalence: a segment ends when its agent flips, and an agent whose confidence is falling is more likely to flip, so negative slopes are censored and p_{SR} should be read as an upper estimate.

At the task level the marginal relationship between $\bar{\beta}$ and accuracy is near zero on GPQA (Pearson $r = 0.02$ fully-connected, $r = 0.04$ star) and weakly negative on HiddenBench ($r = -0.21$ and $r = -0.31$). As discussed in Section 6.2.1, this marginal figure hides a cancellation between opposite effects in the GT-present and GT-absent subpopulations, which is why self-reinforcement is reported conditioned on the opening composition rather than marginally.

Condition	Group	n_{segs}	$\bar{\beta}$	p_{SR}
Ceiling	No ceiling	115,137	0.404	0.798
	Ceiling hit	18,559	0.564	0.910
Terminality	Non-terminal	14,341	0.262	0.682
	Terminal	119,355	0.445	0.829
Initial correctness	Init wrong majority	116,400	0.410	0.809
	Init correct majority	17,296	0.535	0.840
Vote stability	Some agent flipped	113,660	0.380	0.796
	Fully static rep.	20,036	0.685	0.908
Init vote dist.	4-0-0-0	21,435	0.663	0.897
	3-1-0-0	50,286	0.412	0.824
	2-2-0-0	26,629	0.370	0.791
	2-1-1-0	34,035	0.345	0.767
	1-1-1-1	1,311	0.308	0.656

Table 10: Mean segment slope $\bar{\beta}$ (confidence points per round) and prevalence p_{SR} by subgroup, all segments pooled across configurations ($N = 133,696$). A segment is *terminal* if it runs to the end of its repetition, and hits the *ceiling* if it reaches the top of the 1–10 confidence scale. Pooled segment counts are descriptive; the subgroup orderings are confirmed at the task level.

A.6 Multistability: Label Distribution

Table 11 gives the per-dataset breakdown of the three multistability classes referenced in Section 6.2.2. Of the 600 cells, 23.7% are monostable, 41.5% multistable, and 34.8% stochastic; HiddenBench is far more stochastic than GPQA, that is, its endpoints more often coexist without the opening votes selecting between them.

Neither structural axis moves either metric. Across the twelve configurations mean N_{eff} spans only 1.93 to 2.12 and mean Cramér’s V only 0.33 to 0.50, with no consistent gradient in W or topology, which is the basis for reading multistability as a property of the question rather than of the configuration.

Dataset	Monostable	Multistable	Stochastic	N_{eff} mean	V mean
GPQA ($M = 4$)	86	151	63	1.97	0.45
HiddenBench ($M \in \{3, 4\}$)	56	98	146	2.06	0.35

Table 11: Multistability label distribution across all 600 (task, config) cells ($N_{\text{tasks}} = 50$, $R = 50$). Label counts pool all tasks per benchmark; HiddenBench mixes 46 three-option and 4 four-option tasks. The N_{eff} column is a within-benchmark mean and should be read per option-count stratum, since N_{eff} is bounded by M and is not directly comparable across option counts.

A.7 Dominance: Full Configuration Grid

Table 12 reports mean raw D per cell. Star D is $3.3\text{--}4.3\times$ higher than fully-connected across all cells, but this gap is largely mechanical: the star centre is the sole eligible conversion source for every leaf, so its influence share is structurally inflated and its surrogate null degenerate. Emergent concentration is therefore identified only for the fully-connected topology, as discussed in Section 6.2.3. There the mean standardised excess $\bar{z}$ over the constrained-shuffle null is positive on both benchmarks, and the flagged fraction, the share of repetitions whose observed D falls in the upper tail of its own surrogate distribution at $p < 0.05$, exceeds the 5 % that threshold produces by chance. The margin is modest at the repetition level, which is why the claim in Section 6.2.3 rests on the consistency of a positive $\bar{z}$ across tasks rather than on the flag rate alone.

Dataset	Topology	Mean D			fc excess (pooled W)	
		$W = 1$	$W = 2$	$W = 5$	$\bar{z}$	Flagged
GPQA	fc	0.154	0.154	0.155	+0.53	7.7 %
GPQA	star	0.601	0.601	0.610	—	—
HiddenBench	fc	0.158	0.178	0.179	+0.73	11.2 %
HiddenBench	star	0.540	0.574	0.578	—	—

Table 12: Mean raw Herfindahl concentration D per (dataset, topology, W) cell. $\bar{z}$ is the mean standardised excess of the observed D over its constrained-shuffle null, and *Flagged* the share of repetitions significant at $p < 0.05$ against that null, where 5 % is the chance rate. Both are reported for the fully-connected topology only, since the star null is degenerate. $N_{\text{tasks}} = 50$, $R = 50$.

A.8 Limit Cycles: Detection Funnel and Recurrence Matrices

Table 13 gives the full limit-cycle detection funnel referenced in Section 6.2.4. Because star spokes driven by a common hub are not independent successes, prevalence is assessed at the repetition level. The detection rule combines a $p < 0.05$ threshold with the period guard $2 \leq \hat{P} \leq \lfloor L/2 \rfloor$, so its false-positive floor lies below the nominal 5 %: measured against the shuffle surrogates it is 0.92 % at the agent level and 0.71 % at the system level, against observed conditional rates of 0.074 and 0.083. The excess is modest in absolute terms, which is why the claim rests jointly on it and on visual inspection: Figure 12 shows that the flagged cases carry the checkerboard recurrence signature of a genuine period-2 orbit rather than the solid block a merely persistent sequence would produce.

Level	Total	Fixed points	Candidates	Flagged	p_{LC} (cond.)
Agent-level	120,000	119,355 (99.5 %)	633 (0.5 %)	47	0.074
System-level	30,000	29,609 (98.7 %)	388 (1.3 %)	32	0.083

Table 13: Limit-cycle detection funnel. *Fixed points*: trailing constant segment $k \geq 3$, excluded from the test. *Candidates*: non-converged sequences with $L \geq 4$. *Flagged*: DET above the 95th percentile of 1,000 shuffle surrogates with dominant period $2 \leq \hat{P} \leq \lfloor L/2 \rfloor$. p_{LC} (cond.) is the flagged share among candidates. The fixed-point and candidate counts do not sum exactly to the total because a small residue of sequences are non-converged yet too short to test ($L < 4$): 12 at the agent level and 3 at the system level. These are excluded as untestable.

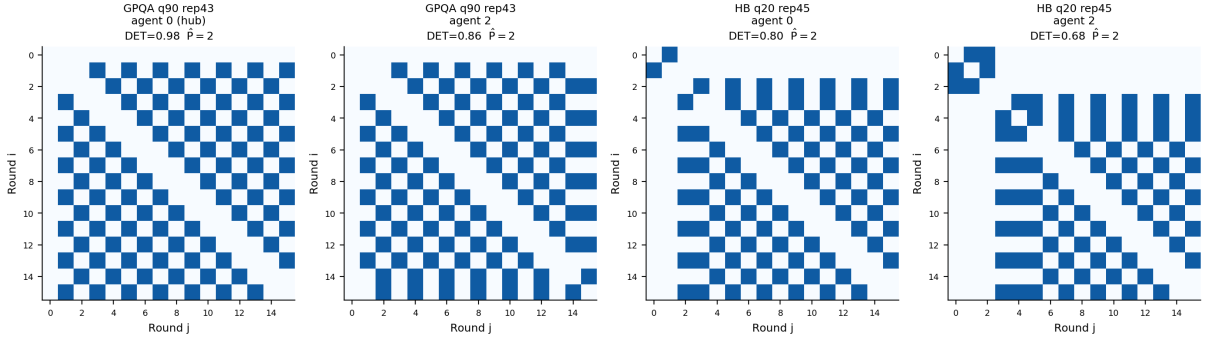


Figure 12: Recurrence matrices for two agents from each of the cases in Figure 8. The alternating checkerboard diagonal stripe pattern (spacing 2) is the characteristic RQA signature of a period-2 orbit; a persistent sequence would instead produce a solid block. DET (determinism) and dominant period $\hat{P}$ are shown above each matrix.

A.9 High-Repetition Analysis: Additional Figures

This appendix provides the feature-by-outcome breakdown for the high-repetition experiment described in Section 6.3 ($R = 1,000$, three GPQA tasks, $W = 2$, fully-connected topology, $N = 4$). The two figures make the same point from complementary directions. Figure 13 shows the conditional means: accuracy rises steeply with initial plurality size (a) but is flat in mean initial confidence (c), mean final confidence (e), and confidence change (f), so none of the confidence channels carries outcome information. Figure 14 shows the same result distributionally: the confidence-change densities for correct and incorrect repetitions overlap almost exactly in all three tasks, whereas the vote-flip and round distributions separate in task-dependent directions.

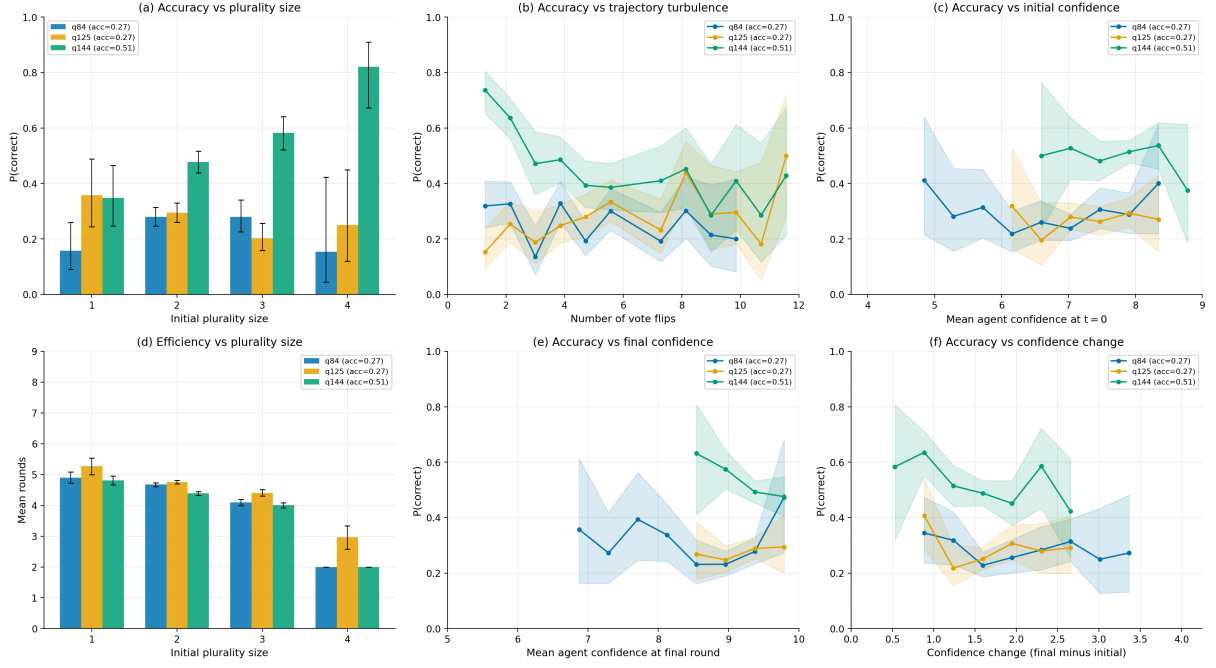


Figure 13: Accuracy and efficiency as a function of six repetition-level features. Top row (a-c): accuracy. Bottom row: efficiency (d) and accuracy (e, f). (a) Accuracy by initial plurality size. (b) Accuracy by vote-flip count (binned, Wilson 95 % CI, $n \geq 10$). (c) Accuracy by mean initial confidence. (d) Mean rounds by initial plurality size. (e) Accuracy by mean final confidence. (f) Accuracy by confidence change (final minus initial); dotted line at zero. $R = 1,000$ per task, $W = 2$, fully-connected.

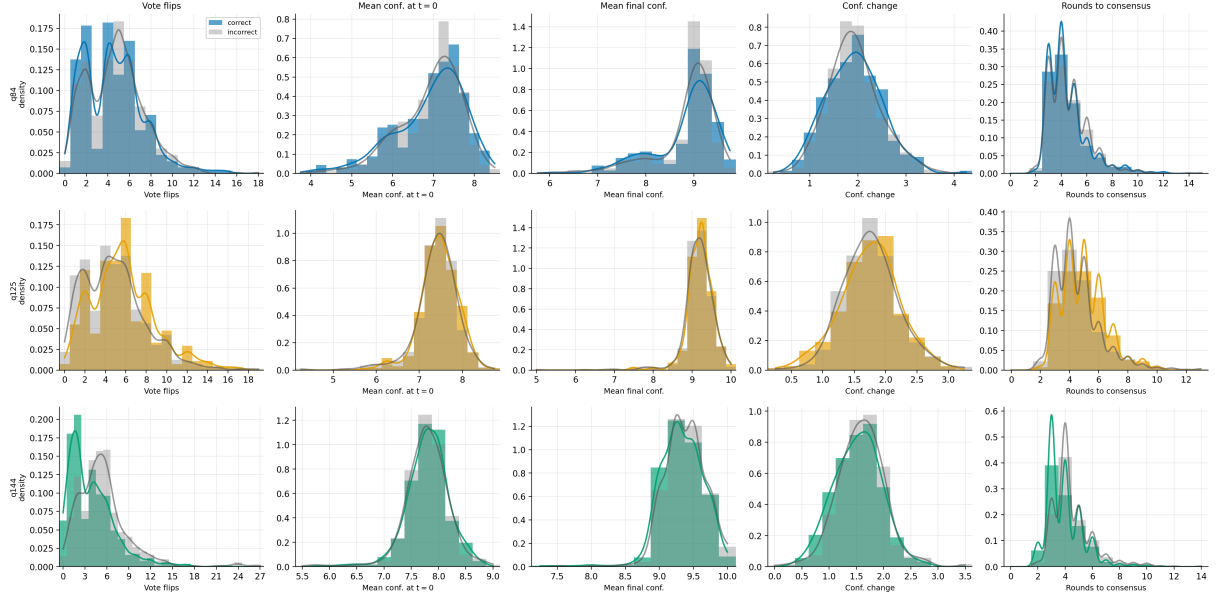


Figure 14: Feature distributions split by outcome (correct: colour; incorrect: grey) for three GPQA tasks, $R = 1,000$ each, $W = 2$, fully-connected. Rows: q84, q125, q144. Columns: vote flips, mean initial confidence, mean final confidence, confidence change (final minus initial), rounds to consensus. Histograms show empirical density; curves are kernel density estimates. The confidence-change distributions overlap almost perfectly in all three tasks, indicating that agents gain confidence at the same rate regardless of whether the system reaches the correct answer.

A.10 Motifs Beyond the Present Selection

Section 5.1 restricts the study to four motifs. The same structural/dynamical lens admits several others that we leave to future work. On the structural side, in the emergent influence graph:

- **Mutual-reinforcement dyad** (a two-node positive loop): two agents that repeatedly push each other the same way.
- **Influence feed-forward loop**: A sways B , B sways C , and A also sways C directly, predicting delayed or robust lock-in.
- **Cascade / chain**: sequential adoption $A \rightarrow B \rightarrow C$, the structural signature of herding.

On the dynamical side, further regimes of the joint trajectory:

- **Consensus / fixed point**: the baseline regime (a point attractor) against which the others are defined.

- **Polarisation / stable split:** persistent within-repetition disagreement into stable subgroups, distinct from across-repetition multistability.
- **Negative autoregulation / self-correction:** the sign-flipped twin of self-reinforcement, an agent that damps its own confidence. Since Section 7.1 identifies the absence of negative feedback as the mechanism behind the amplifier, this is the most immediately consequential of the six.

These span a coherent picture of the regimes a small interacting system can fall into; a study must start somewhere, so we treat them as directions for future investigation.

B Electronic appendix

The complete source code, including all scripts required to reproduce the figures and results, is publicly available in the following GitHub repository:

<https://github.com/echtermeyer/LLM-Based-MAS>

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AI Tools

I hereby confirm that I have written all parts of this work myself. The following AI tools were used during the creation of this thesis and the presentations. All AI-generated suggestions and outputs were critically reviewed, tested where applicable, and independently revised by me. No AI tool was used for the main analysis or interpretation of experimental results.

- **GitHub Copilot** (Inline Completion): Used for inline code completion suggestions during development. All outputs were reviewed and modified as appropriate.
- **Anthropic Claude Sonnet 5** (AI as Tutor): Used to explore and understand core concepts in complex-systems theory and multi-agent debate.
- **Anthropic Claude Opus 5** (Performance and Resource Optimization): Used to parallelize the evaluation pipeline for the experiments described in Section 4 and reported in Section 6. The generated code was reviewed and tested carefully.
- **Anthropic Claude Opus 5** (Improve Writing Style, Paraphrasing and Reformulation): Used for proofreading, linguistic refinement, and paraphrasing. Suggestions were not adopted verbatim but were manually reviewed, adapted, and reformulated.

All AI outputs and suggestions were independently evaluated and selectively adopted after critical review.

München, 04.10.2026

Eric Echtermeyer

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